

PC CONNECTOR (INTERFACES.PC.LIVE)

Pin list (left):

- 1: +24V
- 2: +24V
- 3: GND
- 4: GND
- 5: +5V
- 6: +5V
- 7: +3V3
- 8: GND
- 9: PWM1
- 10: PWM2
- 11: PWM3
- 12: PWM4
- 13: GND
- 14: PWM5
- 15: PWM6
- 16: PWM7
- 17: PWM8
- 18: GND
- 19: PWM9
- 20: PWM10
- 21: PWM11
- 22: PWM12
- 23: GND
- 24: GND
- 25: GND
- 26: GND
- 27: EN
- 28: GND
- 29: EN
- 30: ELT.N
- 31: RDY
- 32: BIAS.EN
- 33: K.A
- 34: K.B
- 35: K.FRE
- 36: HOLD
- 37: IMB-SW1
- 38: IMB-SW2
- 39: IMB-SW1
- 40: GND
- 41: IL1
- 42: IL2
- 43: IL3
- 44: IL4
- 45: AGND
- 46: A.IH
- 47: B.IH
- 48: A.VK.0
- 49: B.VK.0
- 50: AGND
- 51: A.VK.0
- 52: B.VK.0
- 53: IMB-AUX.0
- 54: AGND
- 55: NTC1
- 56: NTC2
- 57: NTC3
- 58: NTC4
- 59: NTC5
- 60: NTC6
- 61: NTC7
- 62: NTC8
- 63: NTC9
- 64: NTC10

Power connections (right):

- +24V to GND via C101 (10u 50V)
- +24V to GND via C102 (100n 50V)
- +5V to GND via C103 (10u 16V)
- +5V to GND via C104 (100n 50V)
- +3V3 to GND via C105 (10u 16V)
- +3V3 to GND via C106 (100n 50V)

X9555W-2x32-6TV01

Both boards live: GND = AGND = BUS-. Supplies +24V / +5V / +3V3 go TO the control board. Commands are 3.3 V logic after the control board's trip latch: every command input is pulled low here (open = off).
 PV-P75: PWM13-56, NTC4 and NTC5 not connected; IL4 pulled to AGND (0 V = no phase 4).

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 PV-P75: PWM13-16, NTC4 and NTC6 not connected; IL4 pulled to AGND (0 V = no phase 4).

COMMAND INPUTS: PULL-DOWNS AND 3.3 → 5 V BUFFERS (GDRV.INPUT_BUFFERS)

NSI6651 inputs are specified at VCC1 = 5 V only (VINH ≤ 3.5 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHC1G08 (VIH 2.0 V) per line. Each PWMn is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48–0.8 us).

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RDY (WIRED-AND) AND FLT_N (WIRED-OR)

The schematic shows the connection of the RBY and FLT_N pins to the TPS37100DCR (U116) and the associated pull-up resistors and capacitors. The RBY pin is connected to +5V through a 10k resistor (R117) and a 100pF capacitor (C124). The FLT_N pin is connected to +5V through a 10k resistor (R119) and a 100pF capacitor (C123). The SUP5_SNS pin is connected to +5V through a 10k resistor (R119) and a 100pF capacitor (C123). The RBY pin is also connected to +5V through a 10k resistor (R117) and a 100pF capacitor (C124). The SUP5_SNS pin is also connected to +5V through a 10k resistor (R119) and a 100pF capacitor (C123).

RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (line 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT # port OC: pull-up on the control board.
 +5V supervisor TPS3710 107k/10k: RDY low below 4.52-4.69 V (AHCCT VREF == 4.5 V, bias headroom).

RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC: pull-up on the control board.
+5V supervisor TPS3710 107k/10k: RDY low below 4.52-4.69 V (AHCT VCC >= 4.5 V, bias headroom).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N

U117
SN74LVC1G32BVR

Q101
B5S138BK

Q102
B5S138BK

Q103
B5S138BK

R120
10k

R121
10k

R122
10k

R123
10k

C126
100n 50V

Signals: A_HOLD, B_HOLD, MOV_OK, FLT_N, K_A, K_B, K_PRE

Legend:
HOLD = A_HOLD OR B_HOLD (|I_LPORT| >= 0.97-1.03 kA, contactor held). MOV_OK high only while BOTH varistor monitor loops conduct (each res. open) below -200 V on its port. Port DC (I_{IL} = -387+415 A lean_rhumb) pulls FLT_N low (open drain); the control board's latch trips without firmware (aR=fuse gap, port_spec). K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on port sheets).

HOLD = A_HOLD OR B_HOLD ([I_port] >= 0.97-1.03 kA, contactor held). MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC ([I] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain): the control board's latch trips without firmware (aR=fuse gap, port_spec). K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS

The diagram illustrates the ground star points for two components, NT101 and NT102, and four signal pins.

- NT101:** A single tie connects AGND to GND, labeled "Net tie: single AGND-GND star point".
- NT102:** A single tie connects AGND to GND, labeled "Net tie: single AGND-GND star point".
- Signal Pins:**
 - #FLG102:** PWR_FLAG is +5V.
 - #FLG103:** PWR_FLAG is +3V3.
 - #FLG104:** PWR_FLAG is GND.
 - #FLG105:** PWR_FLAG is AGND.

AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (Controller ground = BUS-). Layout: gate-drive marries decoupled to GND locally.

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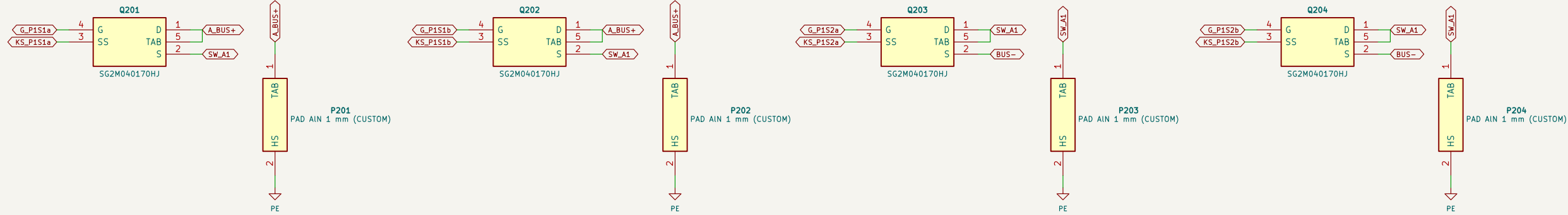
Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 2
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basi

Sheet: /01 PC connector, command buffers, status/
File: 01_pc.kicad_sch

Title: PV-PWR power board – PC connector, command buffers, status

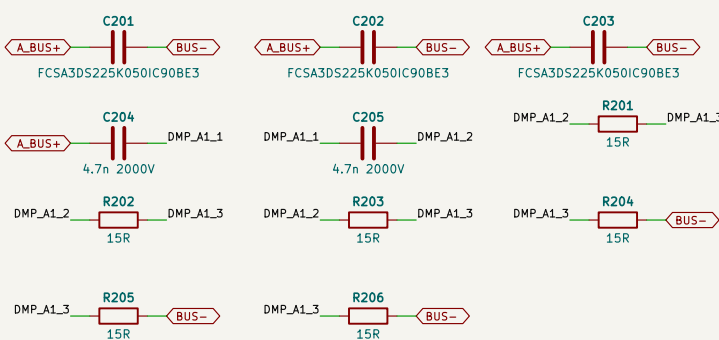
Size: A2	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6		Id: 2/23

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



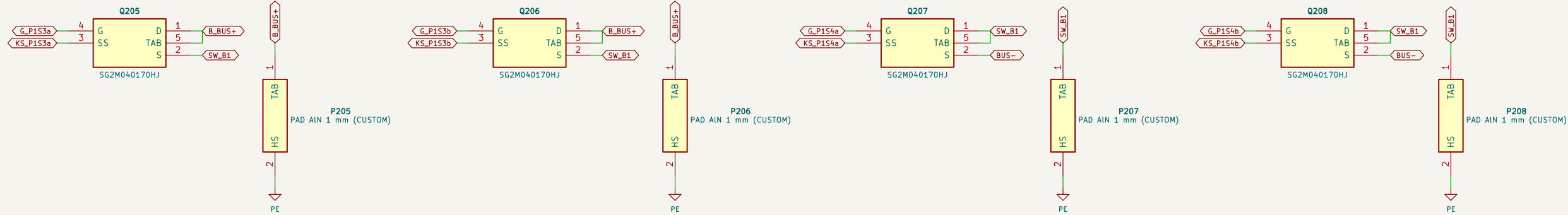
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



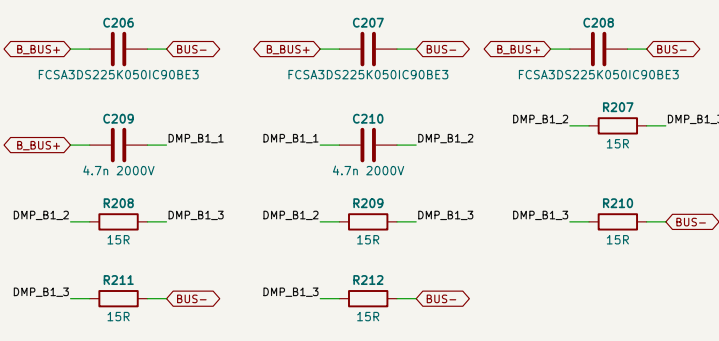
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series; loop at the pins <= 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



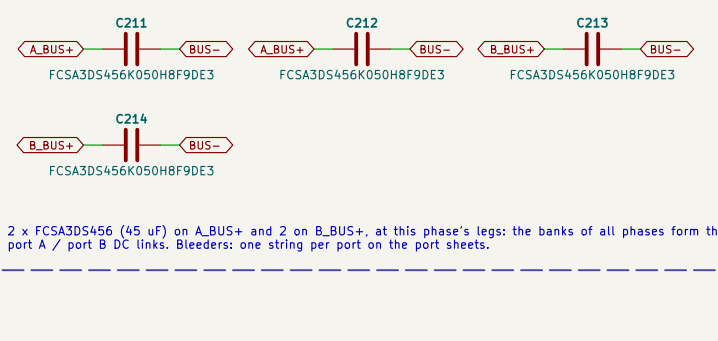
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG B DECOUPLING + RC DAMPER



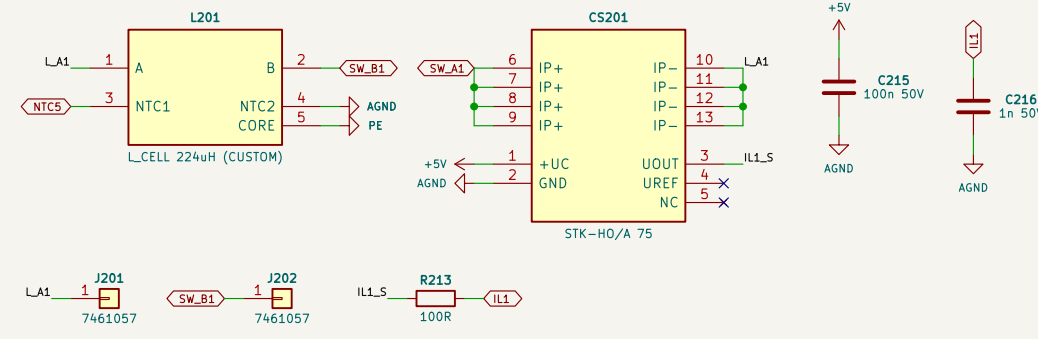
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series; loop at the pins <= 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



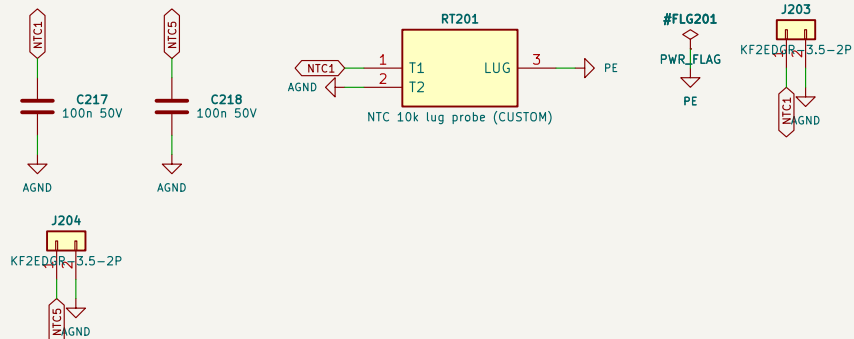
2 x FCSA3D5456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links. Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



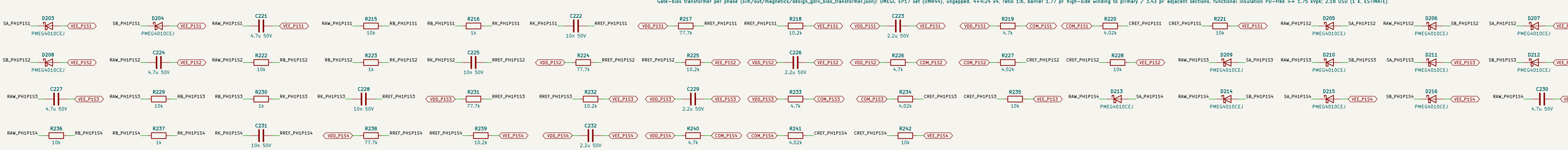
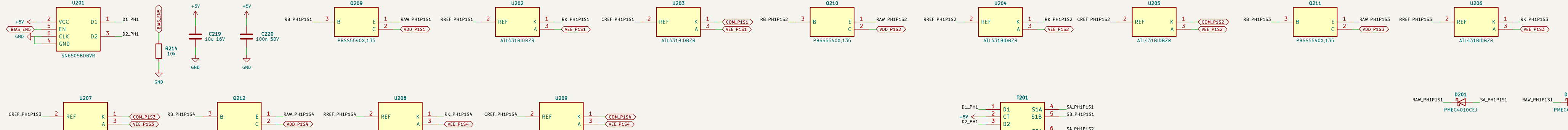
Lead A from SW_A1 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; + = SW_A -> SW_B) to stud
L_A1; lead B on stud SW_B1; IL1 = 2.5 V + 10.6mV/mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A;
100R + 1 nF at the source. Primary at the switch node; functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC1 = heatsink section of phase 1; lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC2 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH1: GATE BIAS (PER PHASE)



SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom; functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

[illegible]

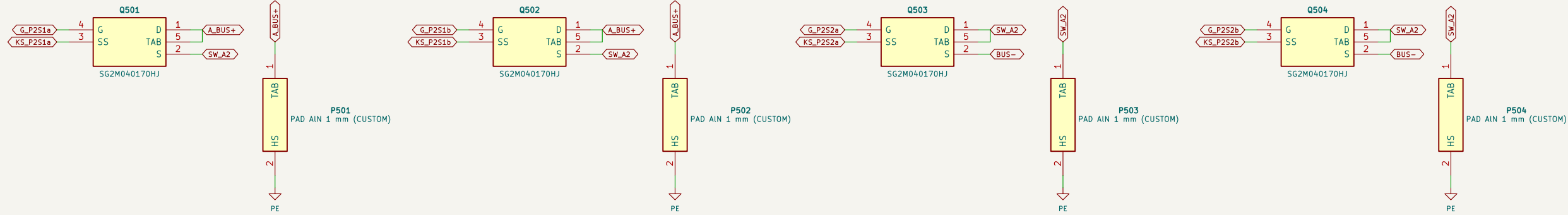
Figure 1: Schematic diagrams of the 16 channels of the 16-channel multiplexed microfluidic system. The top row shows two 8-channel multiplexers (Q1B_P153 and Q1R_P153) controlled by CCG_P153, with outputs Q404 and Q405. The middle row shows two 8-channel demultiplexers (R421 and R423) controlled by VCG_P153, with outputs R422 and R424. The bottom row shows two 8-channel demultiplexers (D408 and D409) controlled by VCG_P153, with outputs D410 and D411. The bottom row also shows two 8-channel multiplexers (R425 and R426) controlled by VCG_P153, with outputs R427 and R428. The bottom row also shows two 8-channel demultiplexers (D408 and D409) controlled by VCG_P153, with outputs D410 and D411. The bottom row also shows two 8-channel multiplexers (R425 and R426) controlled by VCG_P153, with outputs R427 and R428.

[illegible]

Gate and Kelvin of each device as coupled pairs of equal length ($\pm 10\%$) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop ≤ 1 nH, DESAT string ≥ 2.5 mm pad gap per diode: board coated to PD1 (IC-22), Miller hold (ngspice, gdrv); die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

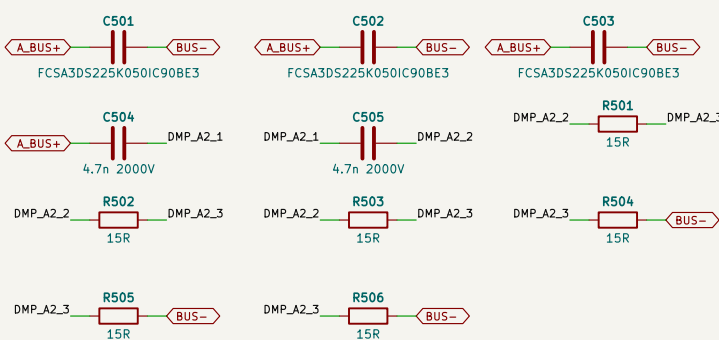
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string <= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 2 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



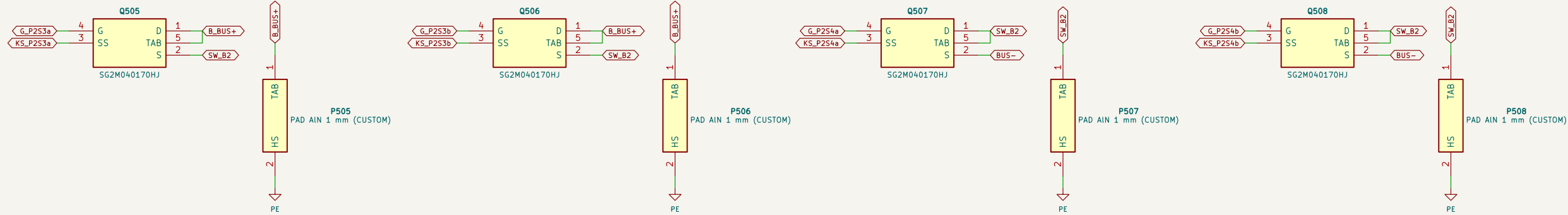
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG A DECOUPLING + RC DAMPER



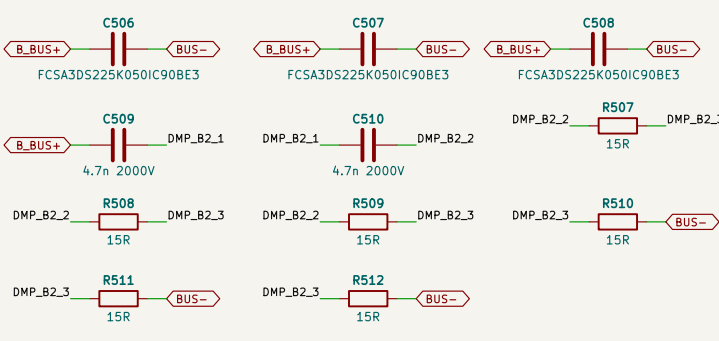
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



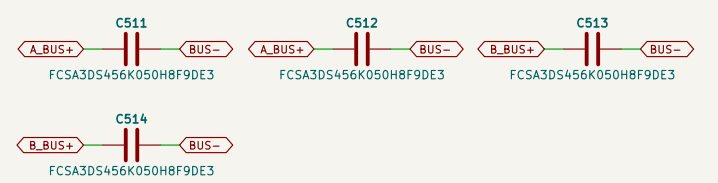
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG B DECOUPLING + RC DAMPER



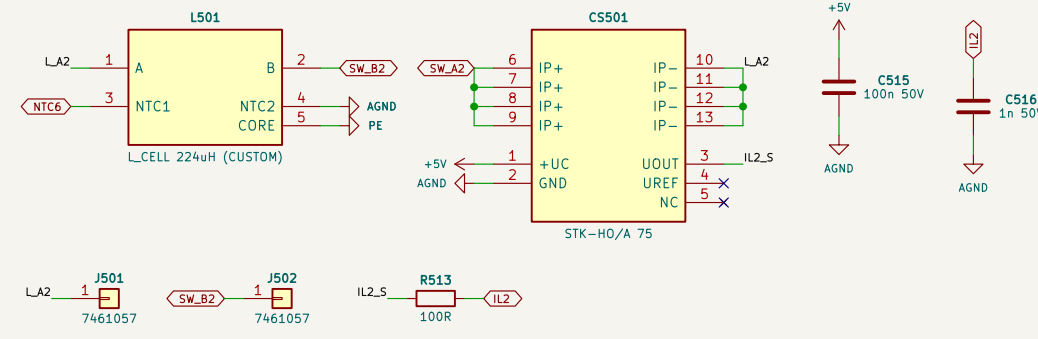
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 2 SHARE OF THE PORT BANKS



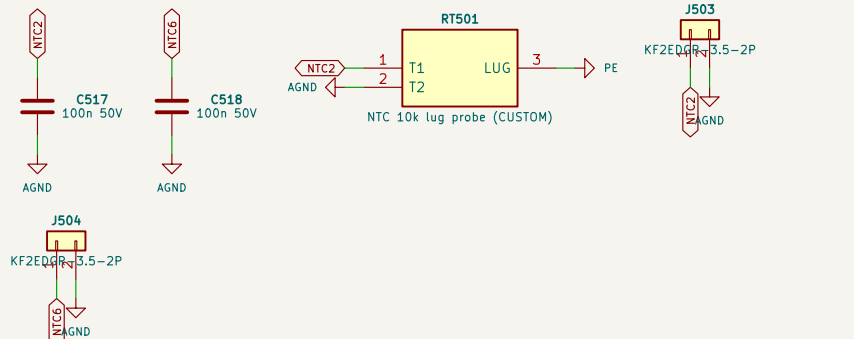
2 x FCSA3DS456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links. Bleeders: one string per port on the port sheets.

PHASE 2 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



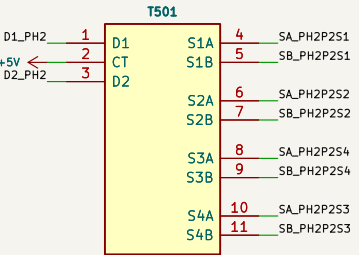
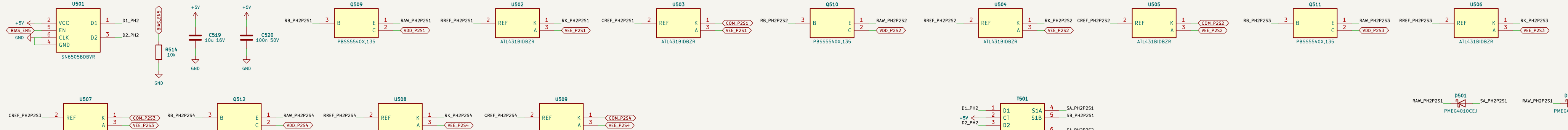
Lead A from SW_A2 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW_A -> SW_B) to stud
LA2; lead B on stud SW_B2; IL2 = 2.5 V + 10.6mV mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A;
100R + 1 nF at the source. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 2 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)

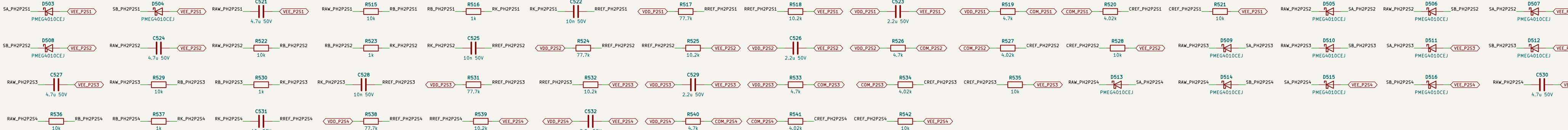


NTC2 = heatsink section of phase 2; lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC0 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH2: GATE BIAS (PER PHASE)

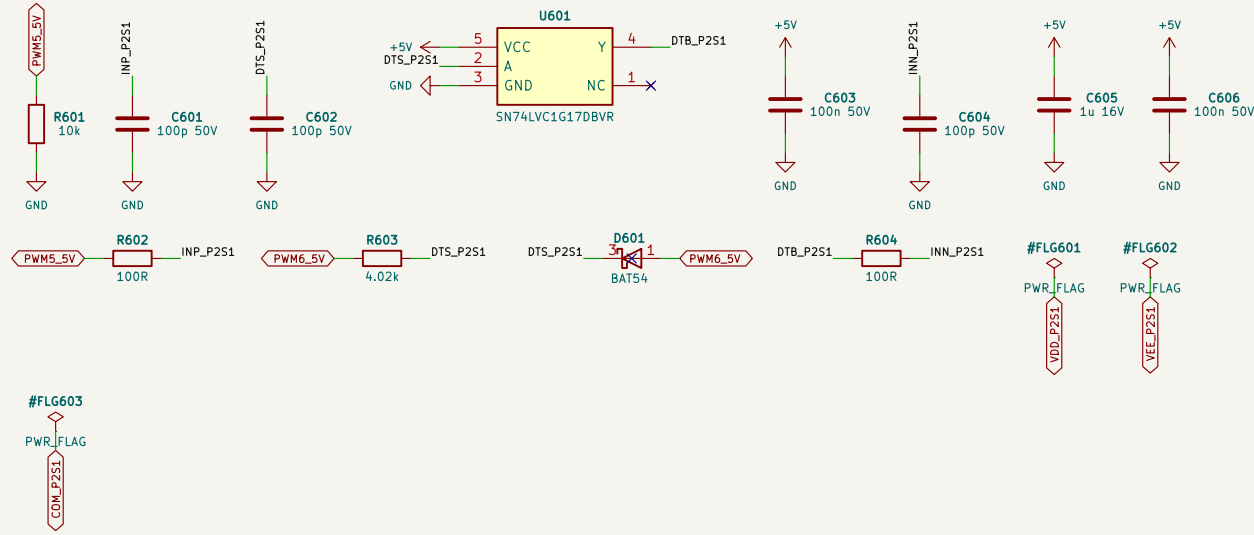


Gate-bias transformer per phase (sim/out/magnetics/design_gdrv_bias_transformer.json): DMEGC EP17 set (DMR44), ungapped, 4+4/24 x4, ratio 1:6, barrier 1.77 pF high-side winding to primary / 3/43 pF adjacent sections, functional insulation PD-free >= 1.75 kVpk; 2.18 USD (1 k, ESTIMATE)



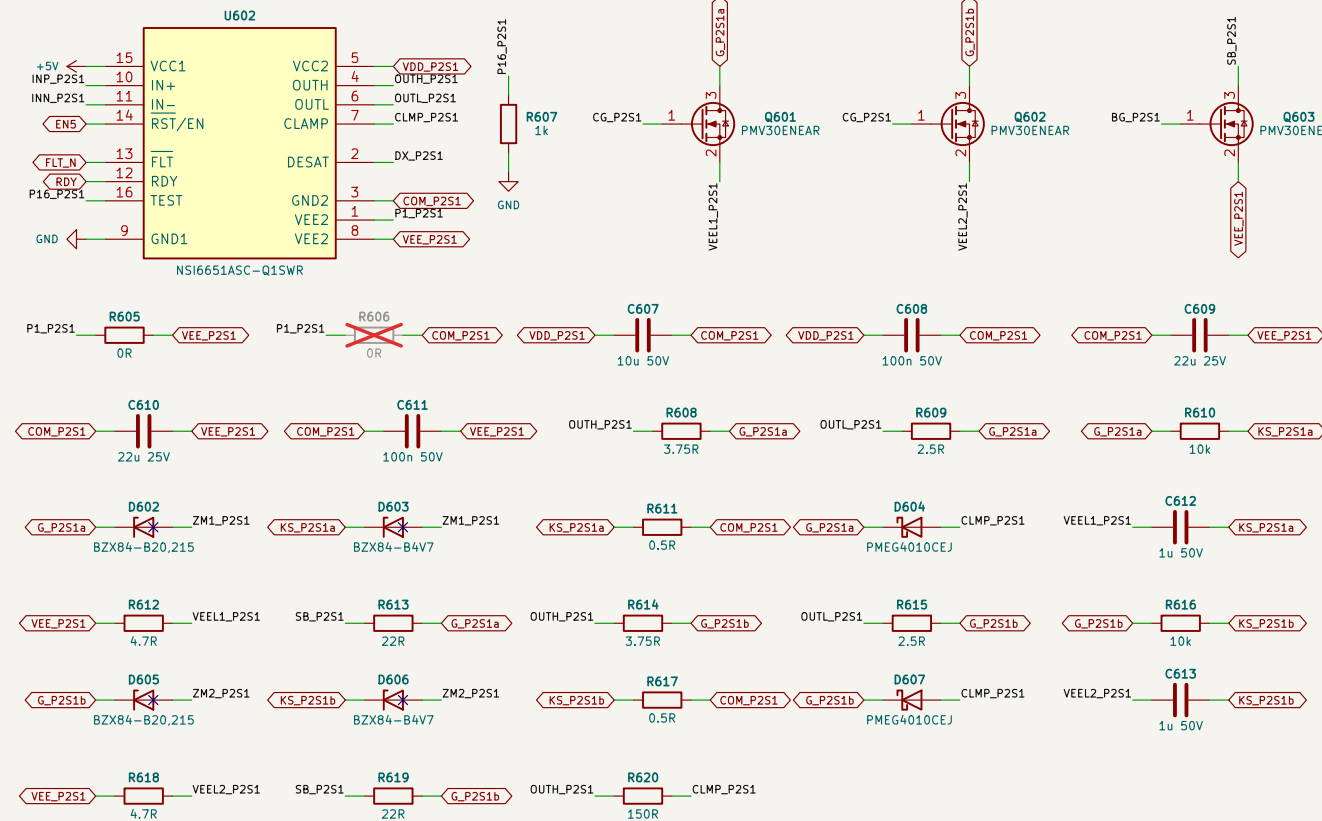
SM6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P2S1: LOGIC SIDE



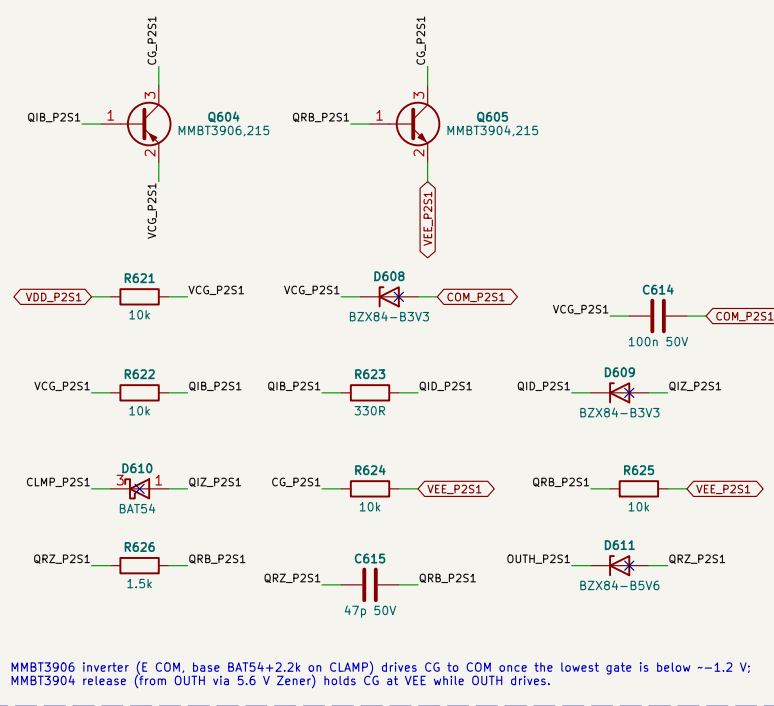
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P2S1: DRIVER AND 2 GATES



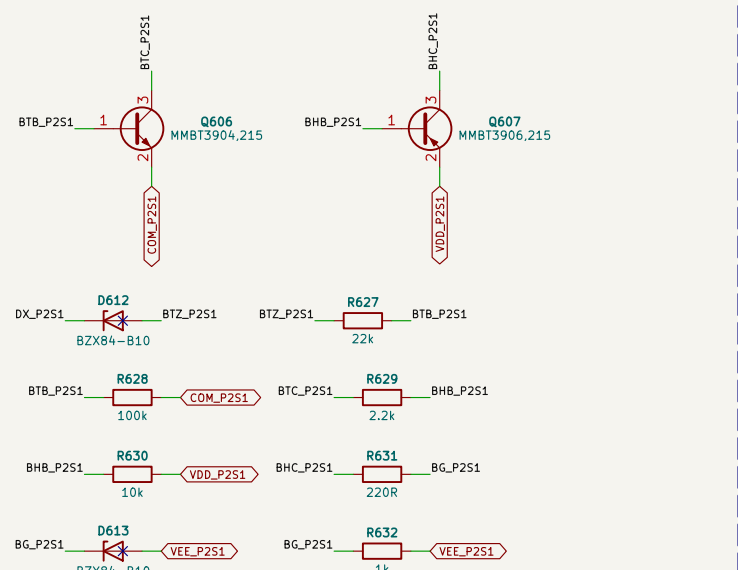
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30E9A) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S1: MILLER CLAMP CONTROL

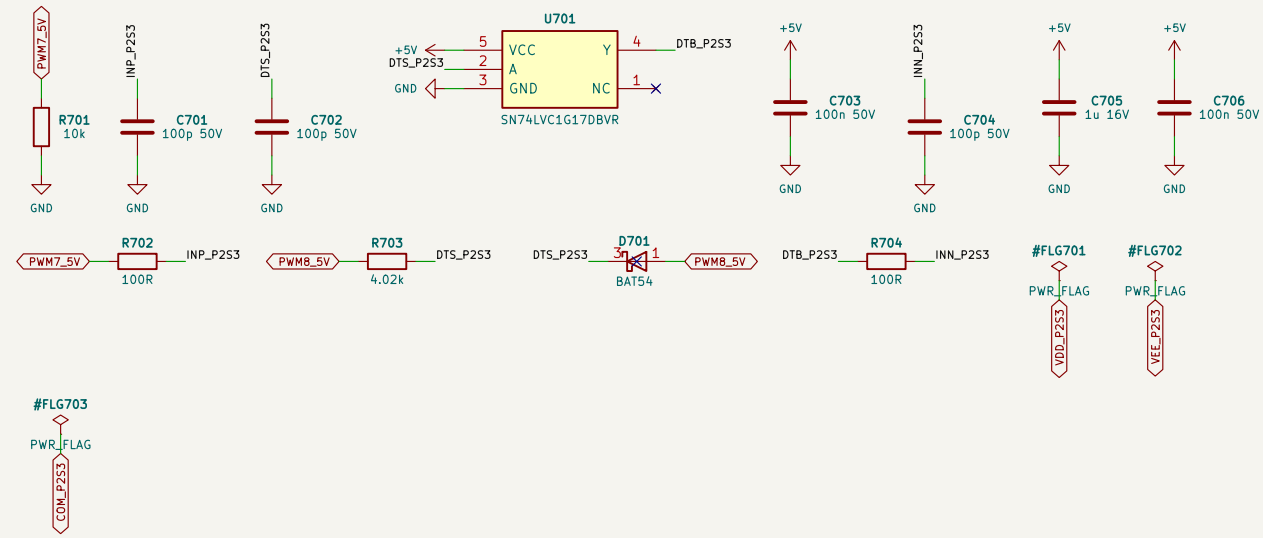


MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S1: SHORT-CIRCUIT BOOSTER

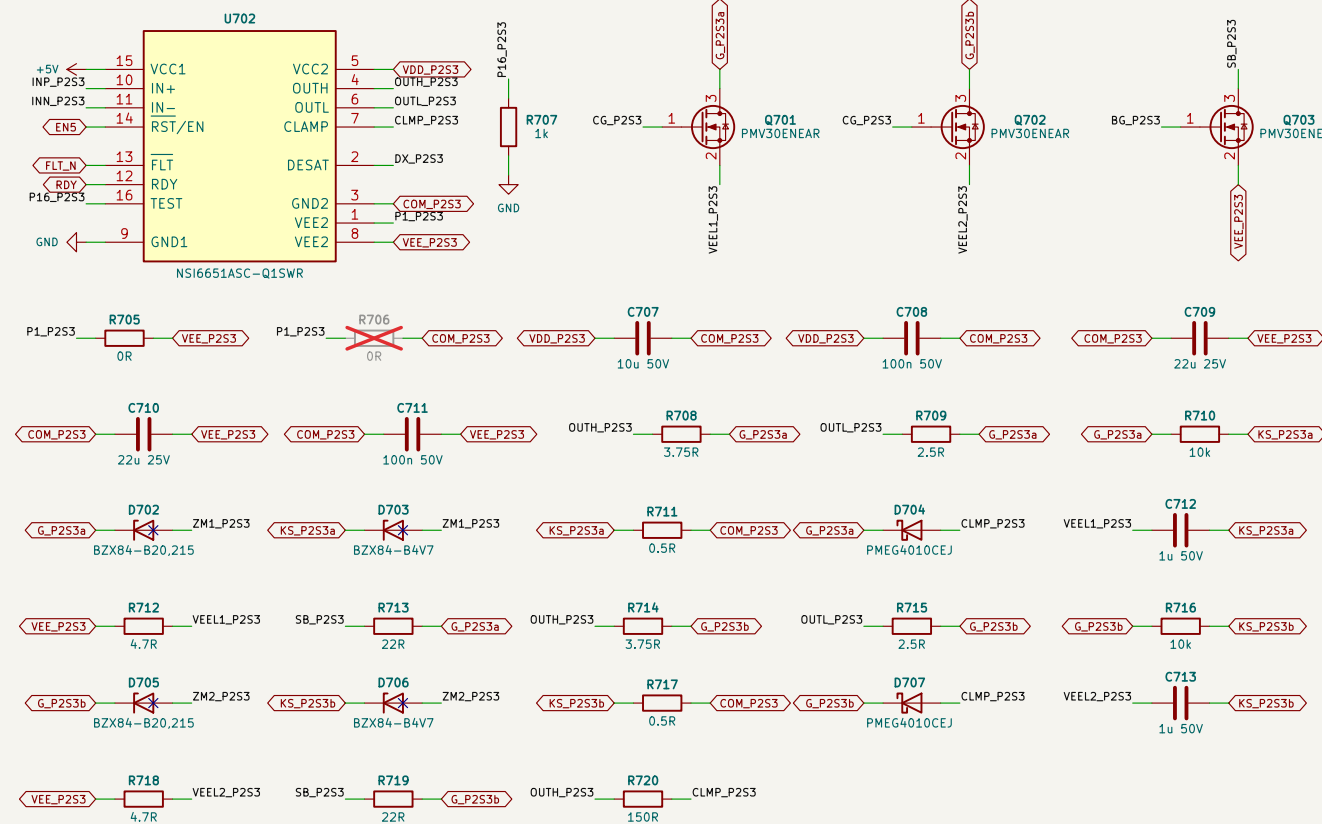


P2S3: LOGIC SIDE



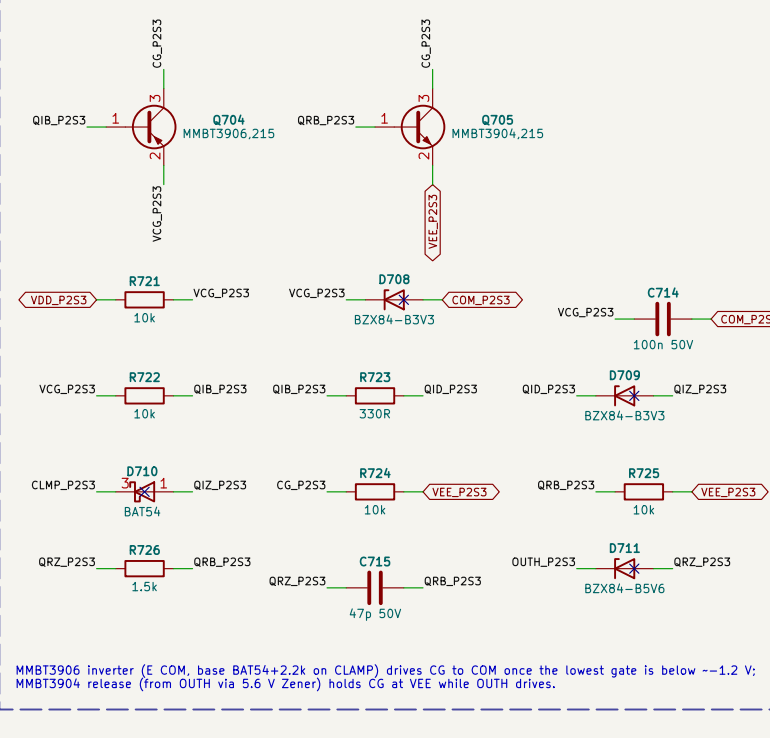
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S3: DRIVER AND 2 GATES



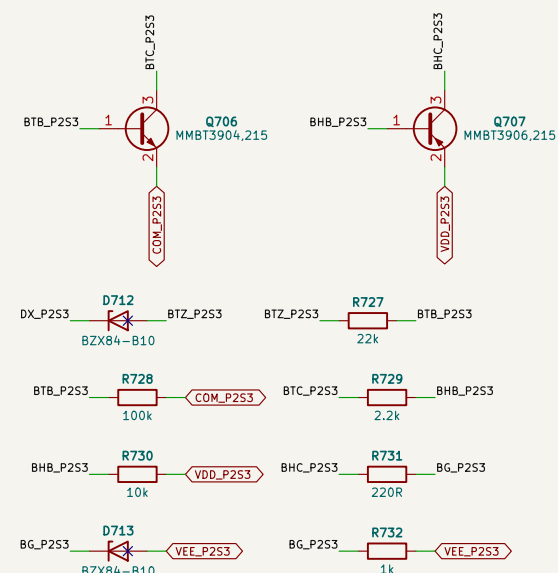
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S3: MILLER CLAMP CONTROL



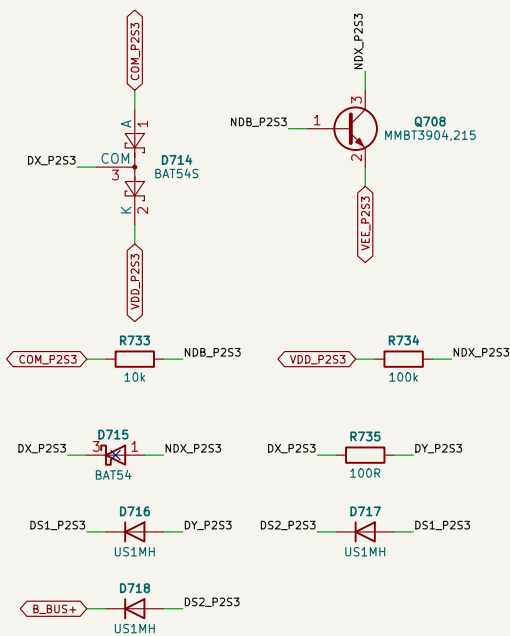
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P2S3: SHORT-CIRCUIT BOOSTER



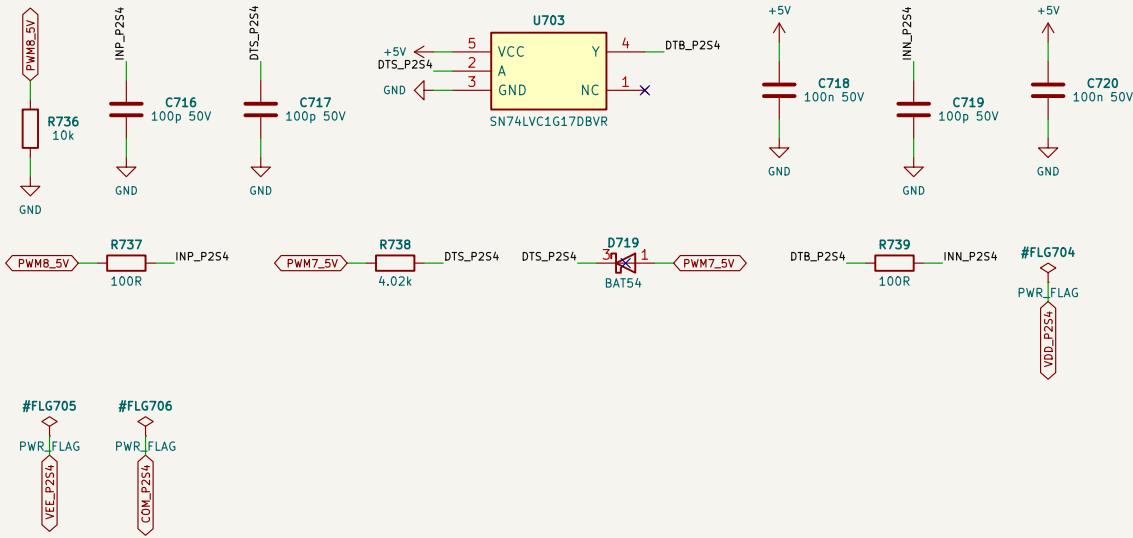
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P2S3: DESAT



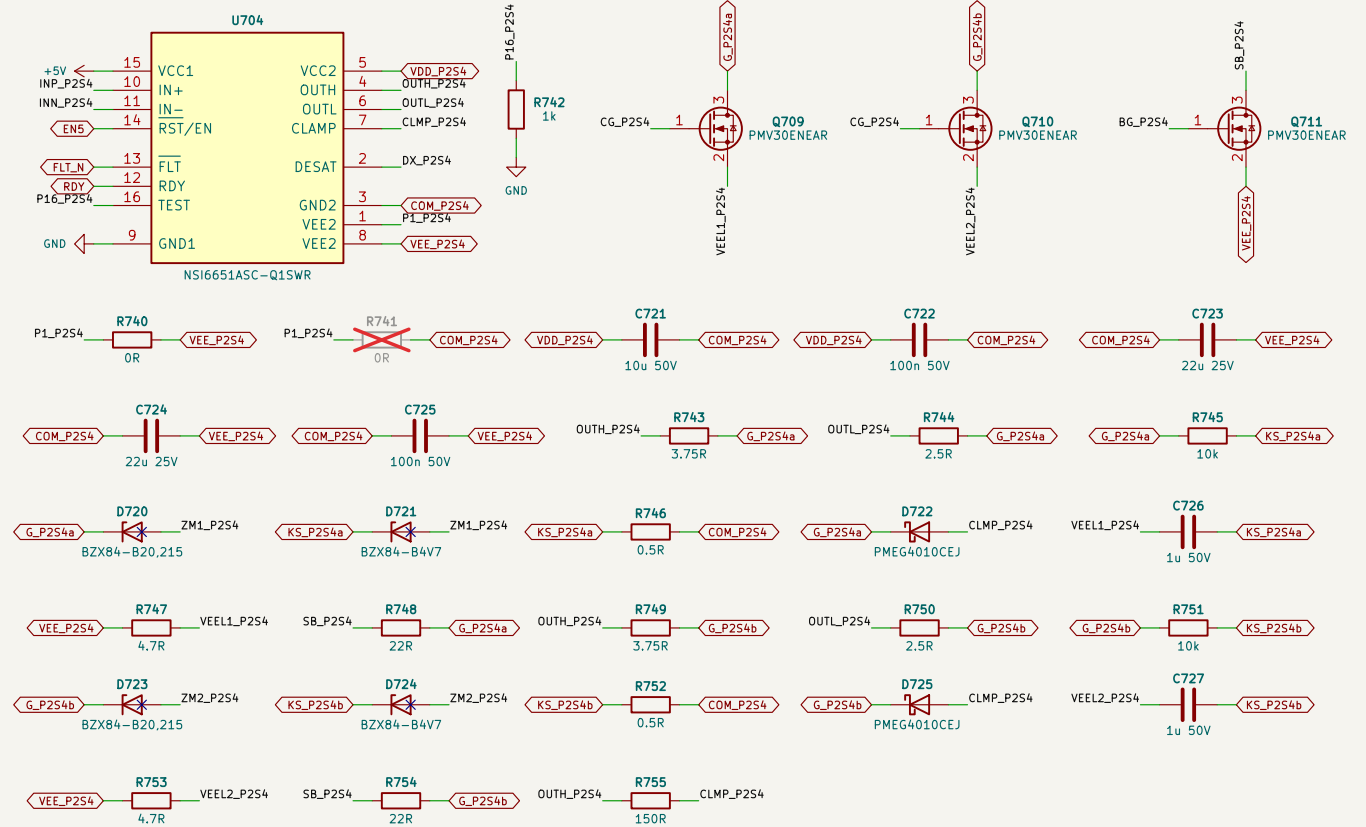
DESAT pin (internal 500 uA blanking source, trip 8.5–9.8 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P2S4: LOGIC SIDE



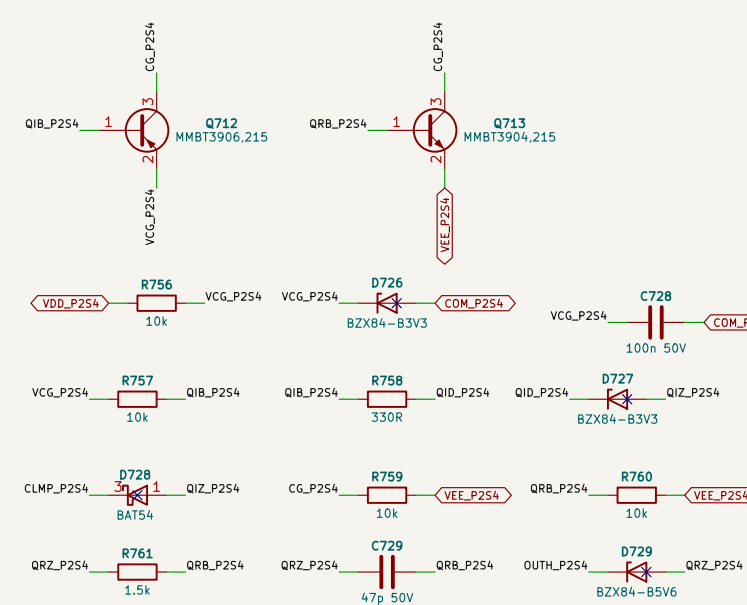
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S4: DRIVER AND 2 GATES



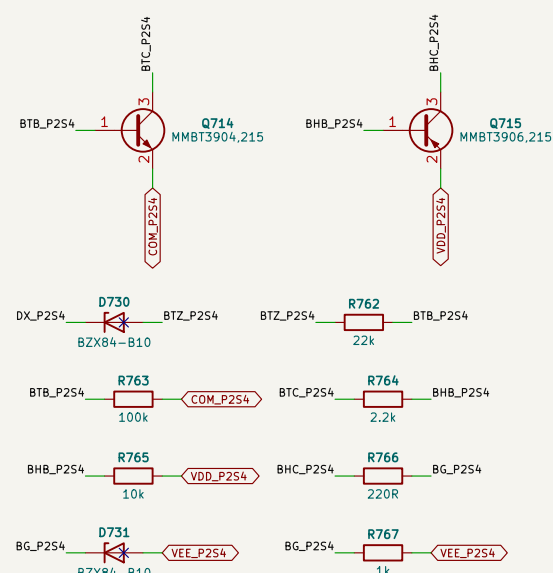
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30NEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S4: MILLER CLAMP CONTROL



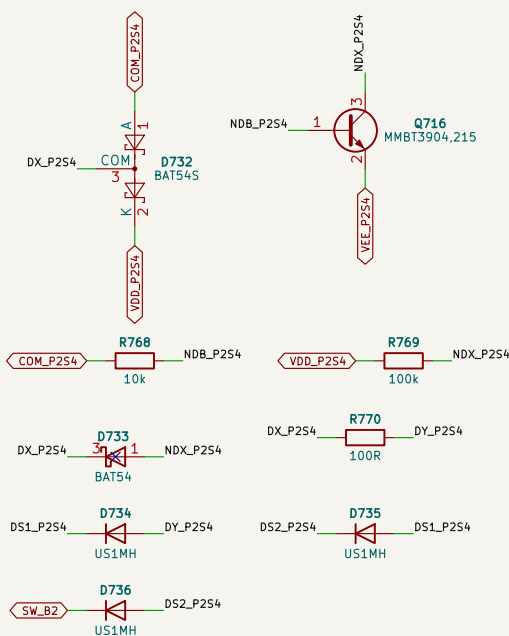
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3–10.9 V (above the 9.8 V maximum trip) → MMBT3904 → MMBT3906 charges BG to VEE + 10 V per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P2S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–9.8 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string: Cblk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 8
Schematics authored by Chinmoy Bhuvan

Sheet: /07 Phase 2 leg B gate drive (S3, S4)/
File: 07_ph2_gdb.kicad_sch

Title: PV-PWR power board – Phase 2 leg B gate drive (S3, S4)

Size: A2 Date: 2026-10-04

Rev: A1

Id: 8/23

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH

The diagram illustrates the RT801 LUG probe setup. It features two NTC thermistors (C817 and C818) connected to AGND. A PE terminal is connected to the LUG probe. A J804 connector is connected to KF2ED06-3.5-2P, which is also connected to AGND. The LUG probe is labeled 'RT801 LUG' and 'NTC 10k lug probe (CUSTOM)'. The connections are numbered 1, 2, and 3.

The schematic diagram illustrates the power supply and I2C interface of the ADXL345 evaluation board. The power supply section (top) features a 5V regulator (U801), a 3.3V regulator (Q809), and a 1.8V regulator (U802). The I2C interface section (bottom) includes an I2C-to-serial converter (U807), a 3.3V regulator (Q812), and an I2C-to-serial converter (U808). The diagram also shows the connection of the ADXL345 (U803) to the I2C interface (TB01) and the connection of the ADXL345 to the I2C interface (D801, D802).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()		
Sheet 9		
Schematics authored by Chinmoy Bhuyan		
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HW-PE bus		
Sheet: /08 Phase 3 power stage, bias, sensing/ File: 08_ph3_power.kicad_sch		
Title: PV-PWR power board – Phase 3 power stage, bias, sensing		
Size: A2	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6		Id: 9/23

The schematic diagram illustrates the NS16651ASC-Q1SWR circuit. The central component is the U902, which is a 16-pin device. Its pins are connected as follows:

- Pin 1:** VCC1 (5V)
- Pin 2:** VCC2 (5V)
- Pin 3:** IN+ (5V)
- Pin 4:** IN- (5V)
- Pin 5:** RST/EN (5V)
- Pin 6:** FLT_N (5V)
- Pin 7:** RDY (5V)
- Pin 8:** P1C_P3S1 (5V)
- Pin 9:** GND
- Pin 10:** VEE1 (5V)
- Pin 11:** VEE2 (5V)
- Pin 12:** DESAT (5V)
- Pin 13:** CLAMP (5V)
- Pin 14:** DX_P3S1 (5V)
- Pin 15:** COM_P3S1 (5V)
- Pin 16:** VEE_P3S1 (5V)

The circuit includes several other components and connections:

- Q901:** PMV30E01, connected to VEE_P3S1 and COM_P3S1.
- Q902:** PMV30E01, connected to VEE_P3S1 and COM_P3S1.
- Q903:** PMV30E01, connected to VEE_P3S1 and COM_P3S1.
- Q904:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q905:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q906:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q907:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q908:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q909:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q910:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q911:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q912:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q913:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q914:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q915:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q916:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q917:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q918:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q919:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.
- Q920:** PMG4010CEJ, connected to VEE_P3S1 and COM_P3S1.

The diagram also shows various capacitors and resistors, including C907, C908, C909, C910, C911, C912, C913, C914, C915, C916, C917, C918, C919, C920, and R907, R908, R909, R910, R911, R912, R913, R914, R915, R916, R917, R918, R919, R920.

The figure displays two schematic diagrams of test boards. The top diagram is for the 100nV test board, and the bottom diagram is for the 100pV test board. Both diagrams show a central component (Q904/Q905) connected to various input and output points. The 100nV board includes components like Q904, Q905, C914, and D909. The 100pV board includes components like D910, D911, C915, and D909. Each diagram also shows the connection to a common ground (COM_P351) and the power supply (VDD_P351).

Figure 1: Schematic representation of the genetic constructs used in this study. The figure shows various plasmid maps and restriction enzyme sites. Top: A vertical plasmid map with sites COM_P351, COM_P351, A, COM, D194, BAT54S, VDD_P351, and VDD_P351. Middle: A circular plasmid map with sites NDB_P351, Q908, MMBT3904_215, and VDD_P351. Bottom: Five linear plasmid maps showing sites like DX_P351, NDX_P351, R933, R914, NDx_P351, D1915, BAT54, R935, 100R, D51_P351, DY_P351, D52_P351, D1917, US1MH, D52_P351, A_BUS+, and US1MH.

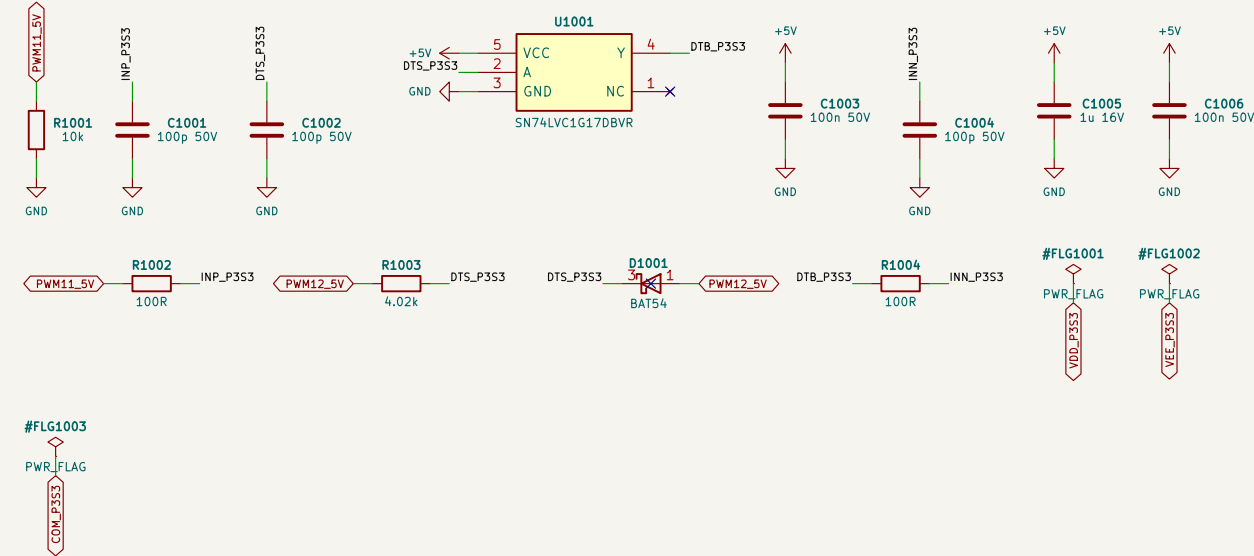
The schematic diagram illustrates the test circuit for the proposed device. It features several key components and connections:

- Power Sources:** Two pulse-width modulated (PWM) voltage sources are used: **PMW10.5V** and **PMW9.5V**.
- Resistors:** Resistors **R936** (10k), **R937** (100R), and **R938** (4.02k) are connected in series with the PWM sources and the DTS_P352 transistor.
- Capacitors:** Two capacitors, **C916** and **C917**, both rated at 100pF and 50V, are connected in parallel with the PNP_P352 and DTS_P352 transistors, respectively, to ground (GND).
- Transistors:** The circuit includes two transistors: **PNP_P352** and **DTS_P352**. The PNP_P352 is connected to the 10.5V source, and the DTS_P352 is connected to the 9.5V source.
- Logic Section:** A logic buffer, **5N74LVCM13**, is connected to the output of the DTS_P352 transistor. Its VCC pin is connected to a 5V supply, and its GND pin is connected to ground.
- Grounding:** All components are referenced to a common ground (GND).

[illegible]

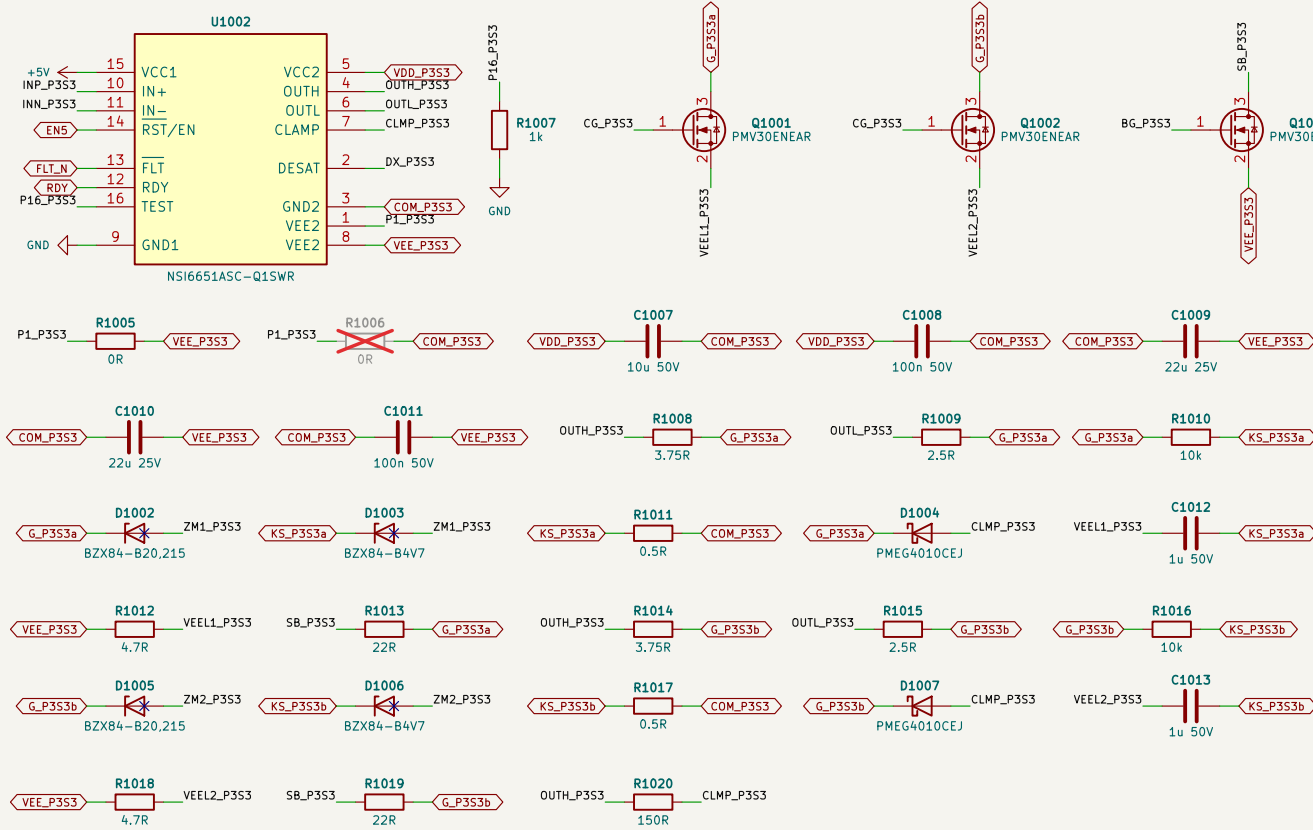
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

P3S3: LOGIC SIDE



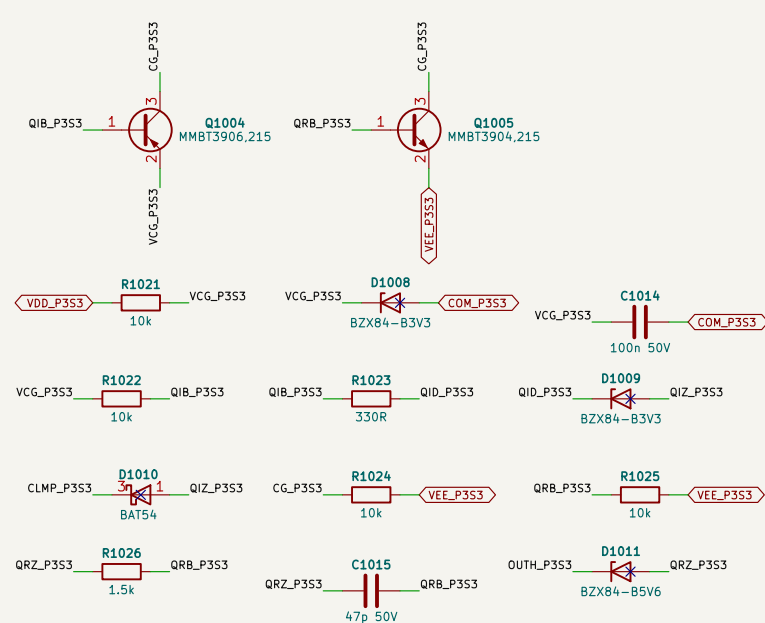
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P3S3: DRIVER AND 2 GATES



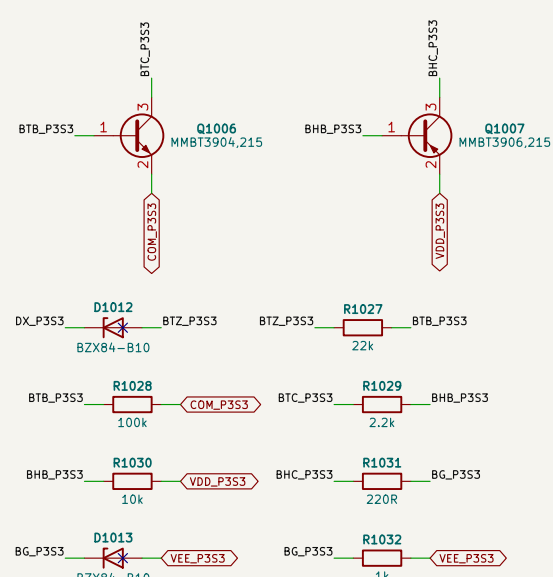
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S3: MILLER CLAMP CONTROL



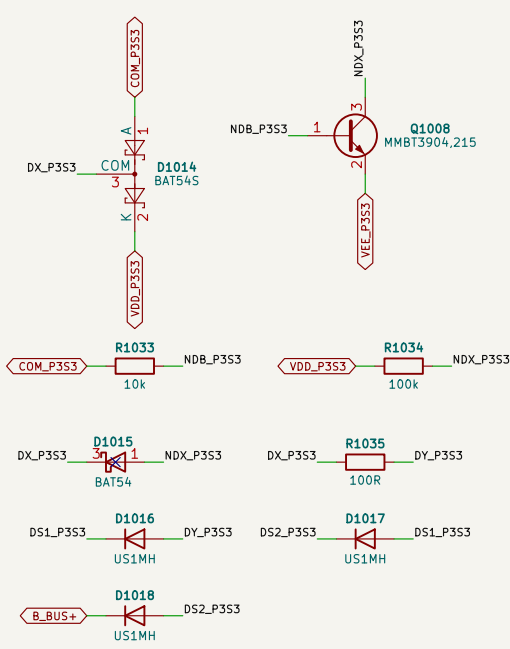
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S3: SHORT-CIRCUIT BOOSTER



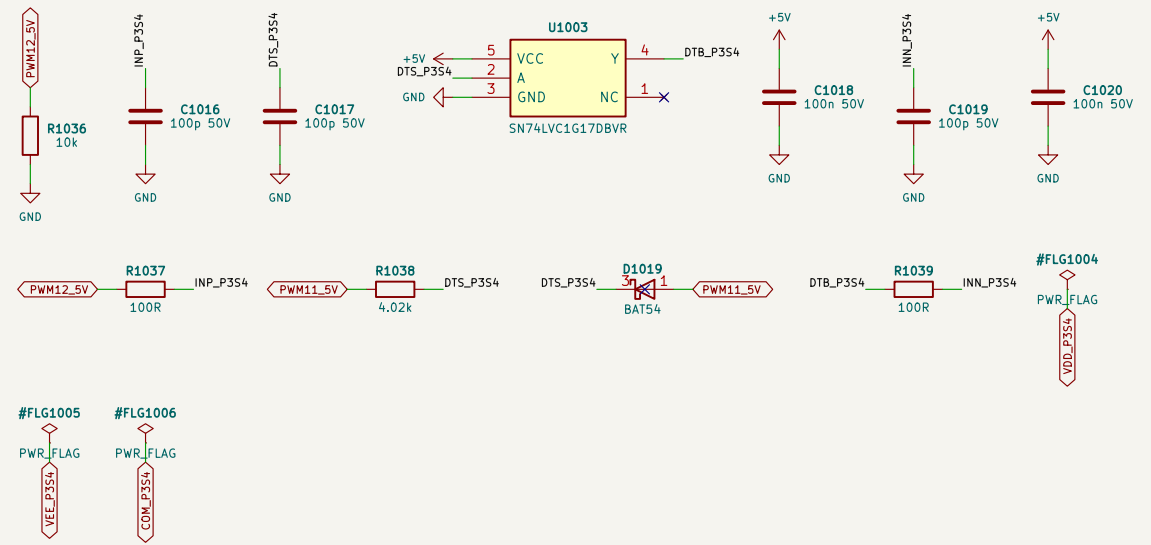
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P3S3: DESAT



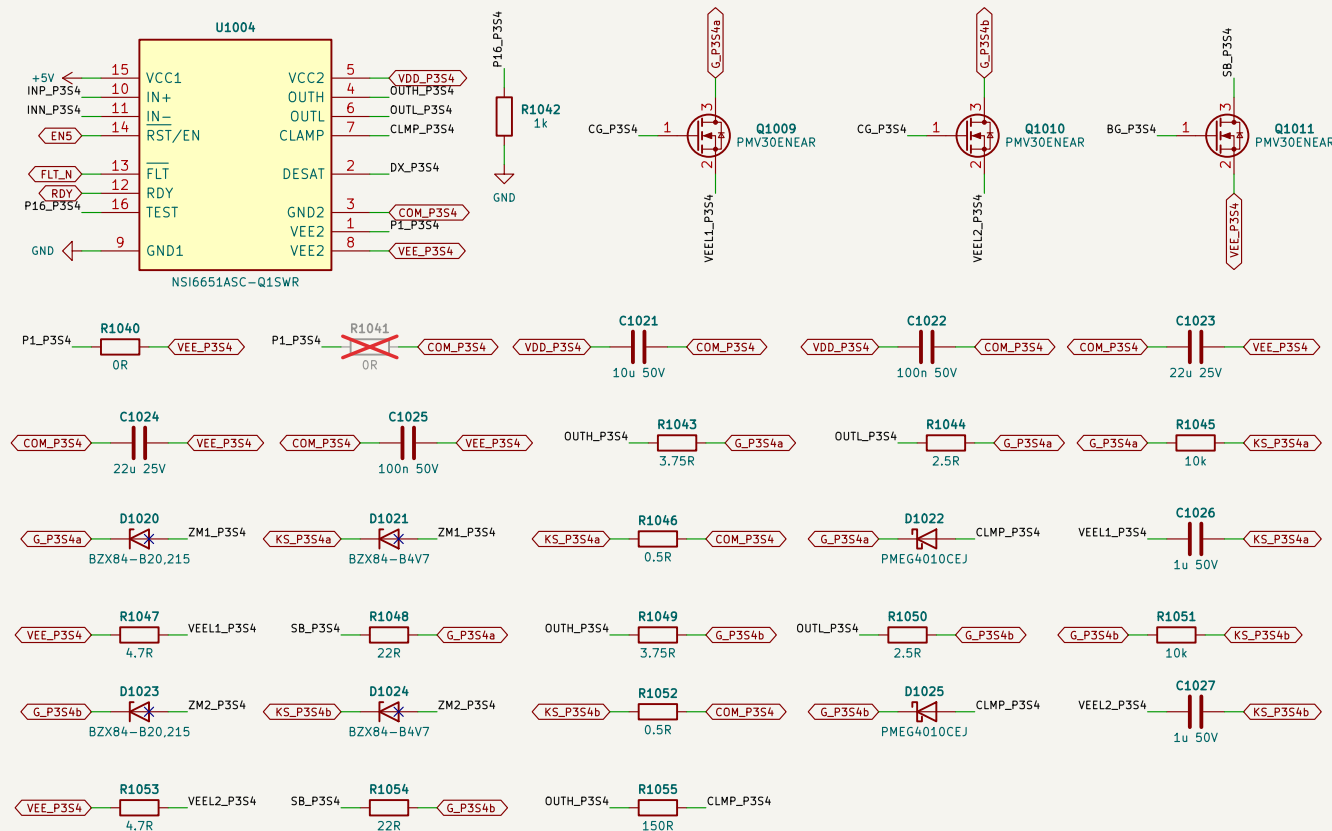
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

P3S4: LOGIC SIDE



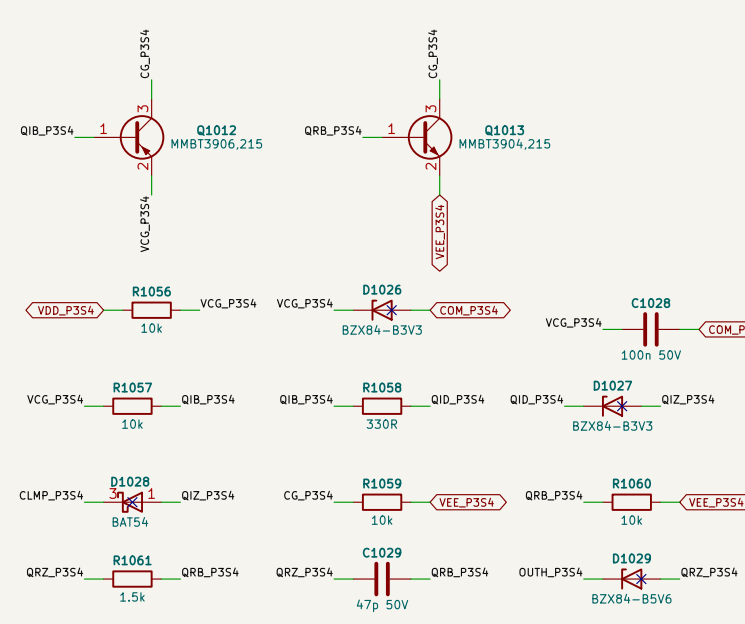
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P3S4: DRIVER AND 2 GATES



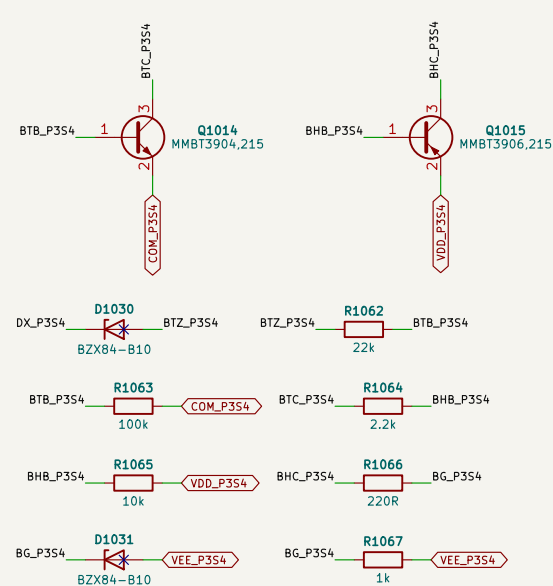
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S4: MILLER CLAMP CONTROL



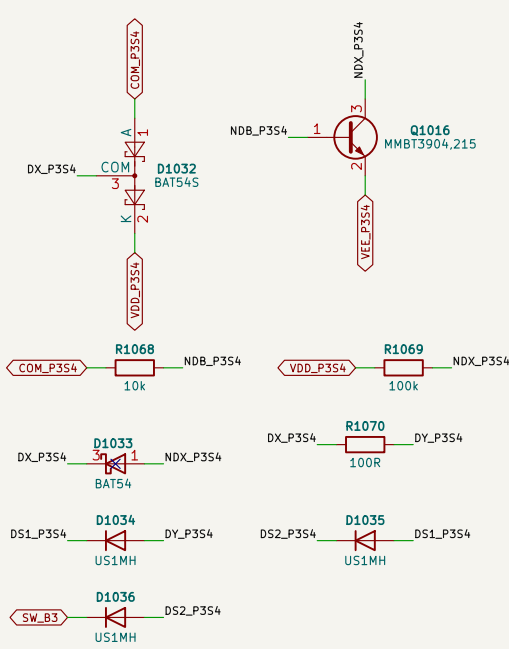
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P3S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + CxV at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22); Miller hold (optical, zero): die +1.41 V vs VGS(n) min 1.94 V at 170 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{LN} / A_{JN}-- bank A_{BUS}-- No fuse, no precharge; array lsc <= 1.25 x rating.
 Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{IN}+ / A_{IN}-, bank A_{BUS}+. No fuse, no precharge; array lsc ≤ 1.25 x rating.
Contactor opens ≤ 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

TV25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE_T. Up. eff 3793 V pole-PE and pole-pole at In 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA. Branch fuses ETI CH14x51 gPV 36 A (30.0 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 12
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /11 Port A power path (pv)/
File: 11_leanA_power.kicad_sch

Title: PV-PWR power board – Port A power path (pv)

Size: A2	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6		Id: 12/23

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 13 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /12 Port A sensing and interlocks/ File: 12_leanA_ctrl.kicad_sch		
Title: PV-PWR power board – Port A sensing and interlocks		
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6	Id: 13/23	

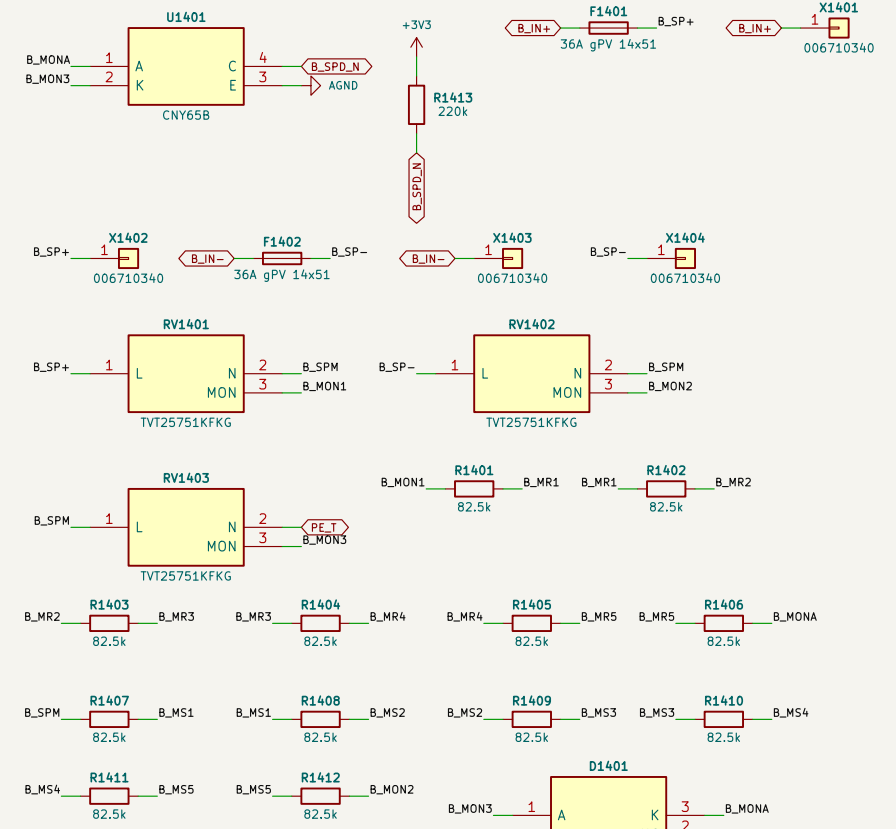
	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 14 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /13 Port A coil drives/ File: 13_leanA_drive.kicad_sch		
Title: PV-PWR power board – Port A coil drives		
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6	Id: 14/23	

PORT B (BATTERY SOURCE), D-044 LEAN PORT

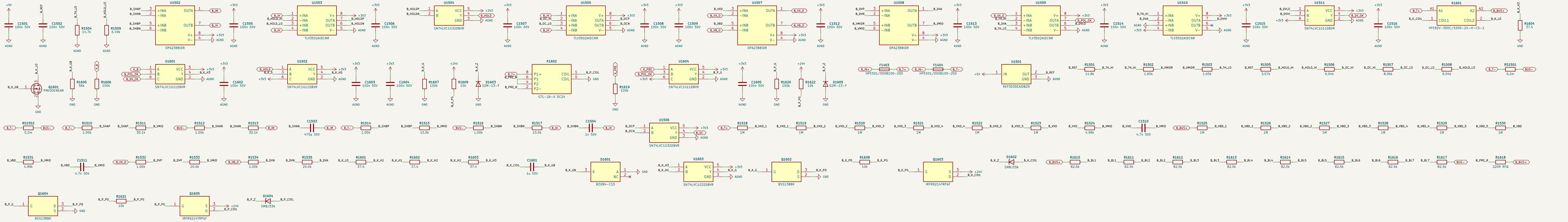
Terminal side R_MN+ / R_MN- bank R_BUS+/- HPE501 250 A at in both poles (breaks 1.25-50 A at in).
Contactor opens <= 0.97 A, held above (port_Lapc time). Installation: upstream circuiters 987-1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TV25751 = 3 (605 VDC, thermal time + monitor load); MOV1 + PM MOV2 - TV MOV3 W/PEL1 Use at 3795 V pole-PE and post-pole at 3 VA (GND, <= 4.0 VV => reinforced 5 VV / basic 4 V, D-032); max 25 mA Branch
From ESI CHANSEL GPF 50 A 50 mA (U/F 2 ms) Monitor loop through 500 Tares and all three lines -> CHANSEL
-> R81 (sum on the ID line); R81 = 1 at rest above 200 V = varistor network open -> CTR1 inhibit/stop.

AR FUSES (BATTERY-TYPE PORT)



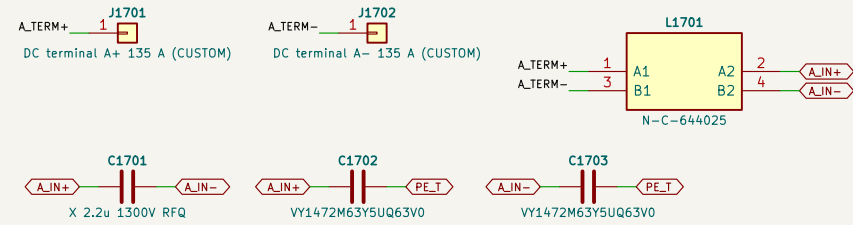
	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 16 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /14 Port B sensing and interlocks/ File: 14_leanB_ctrl.kicad_sch		
Title: PV-PWR power board – Port B sensing and interlocks		
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6	Id: 16/23	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

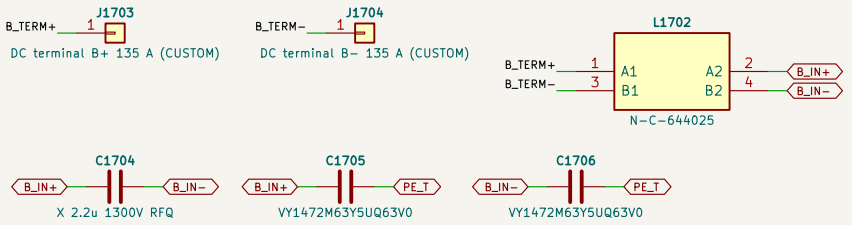
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 17 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /15 Port B coil drives/ File: 15_leanB_drive.kicad_sch		
Title: PV-PWR power board – Port B coil drives		
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6	Id: 17/23	

PORT A TERMINALS AND EMI FILTER



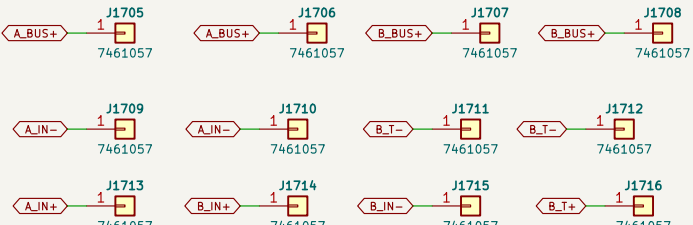
PV: chassis terminals A_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



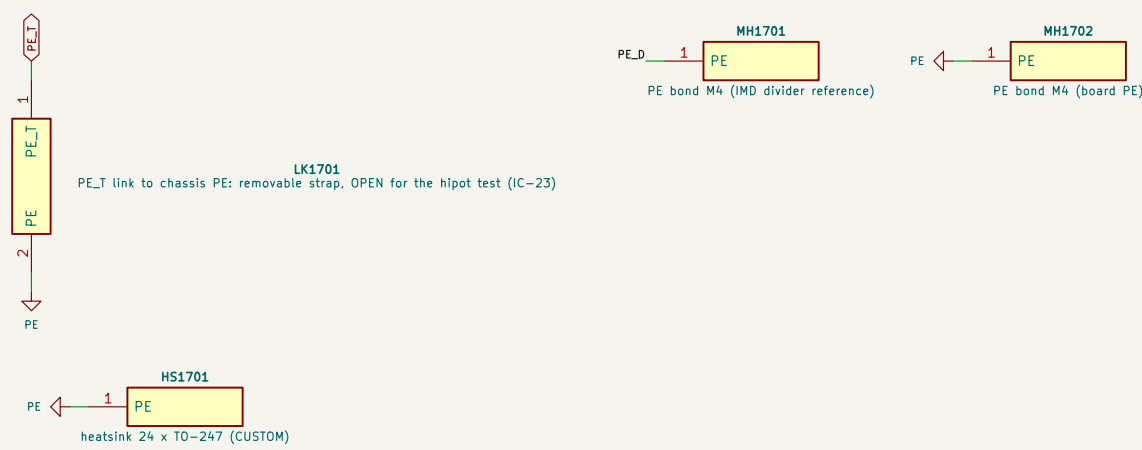
battery: chassis terminals B_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



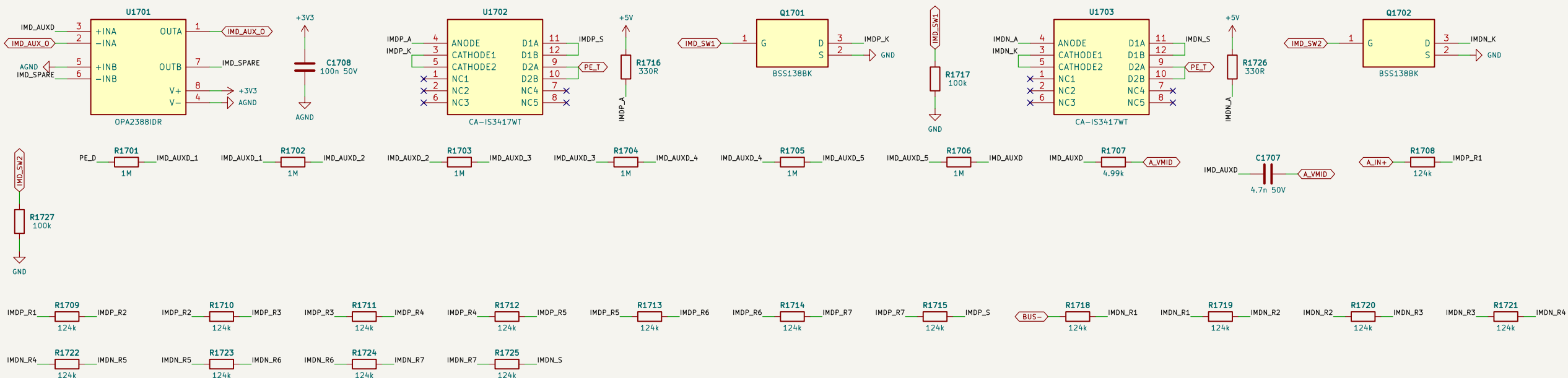
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



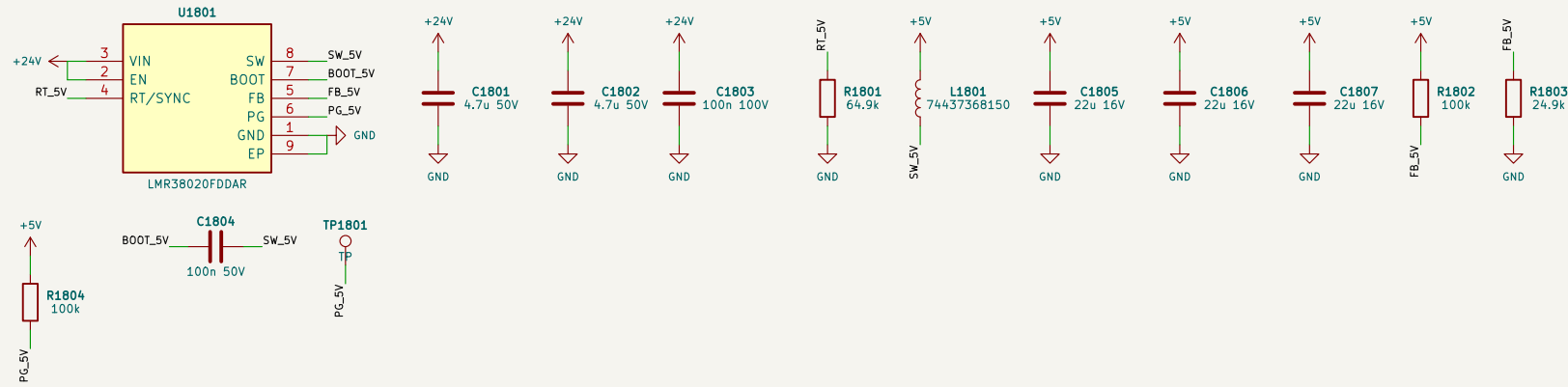
PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



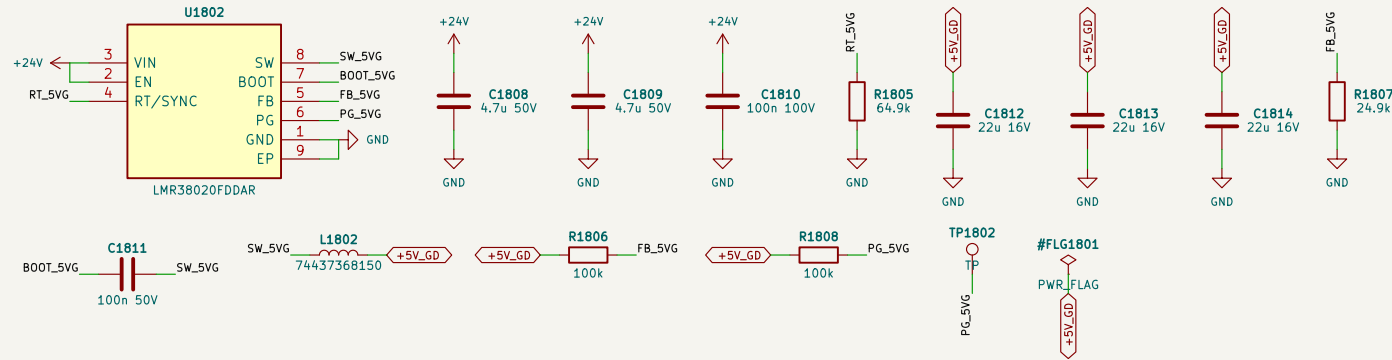
IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected); VPE = PE divider against BUS- (VMID of port A).

24 V -> +5V (LMR38020, 400 KHZ, 2 A)



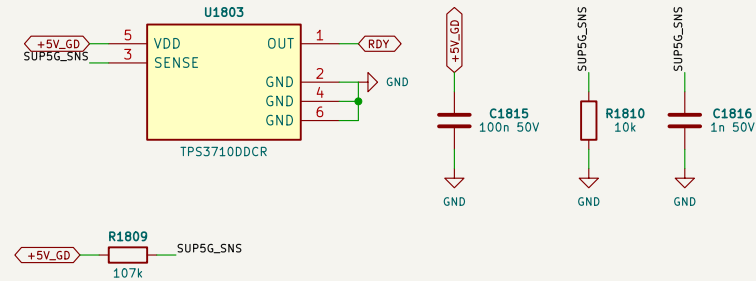
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



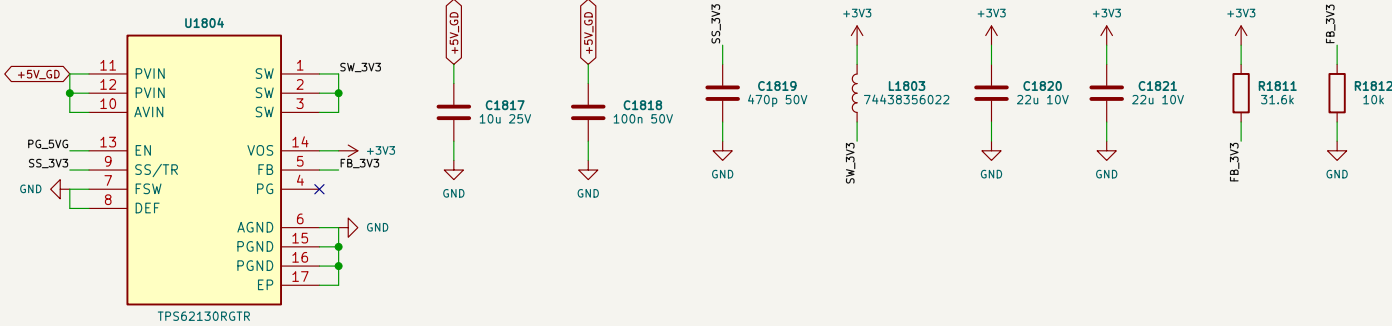
gate bias of phases 3 only, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



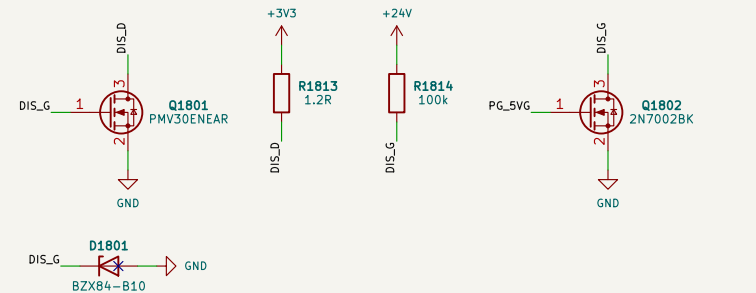
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3 is below 4.52-4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5VG, FB 31.6k / 10k 0.1 % -> 3.240-3.433 V (>= 3.20 V for the control board's reference), SS/TR 470p COG; ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control <= 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5VG low (+5V_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at >= 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3-29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12-36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 19

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Live 5 V / 3.3 V, SELV connector/

File: 16_supplies.kicad_sch

Title: PV-PWR power board - Live 5 V / 3.3 V, SELV connector

Size: A3

Date: 2026-10-04

Rev: A1

KiCad E.D.A. 10.0.6

Id: 19/23

TAP A_IN+



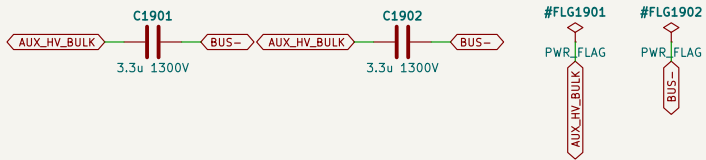
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP B_T+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual -> 1199 V (UR 1300 V); filter corner 546 Hz.

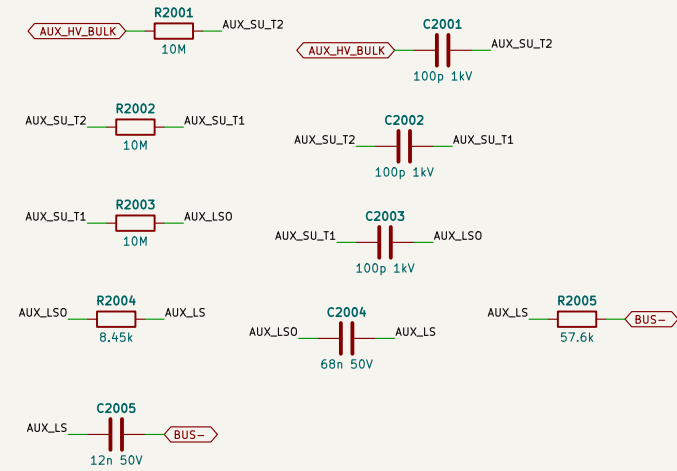
Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 20
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /17 AUX75 input taps, OR, film/
File: 17_aux75_input.kicad_sch

Title: PV-PWR power board – AUX75 input taps, OR, film

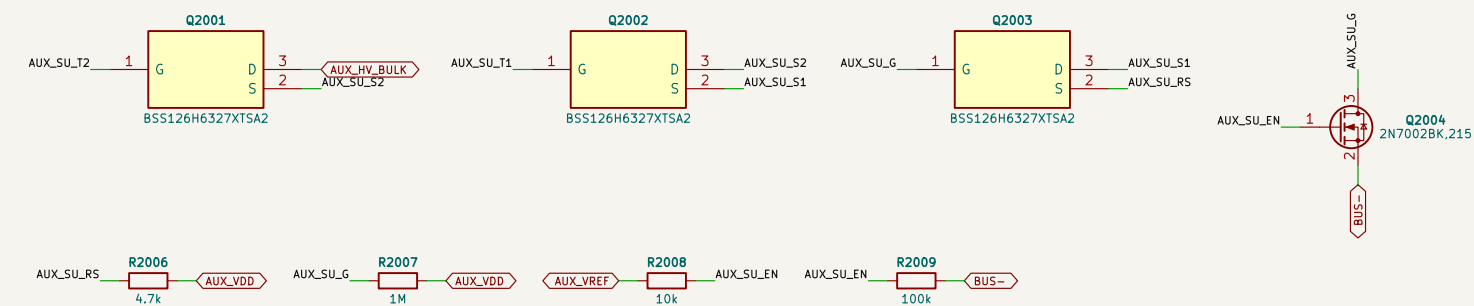
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6		Id: 20/23

COMPENSATED TAP STRING



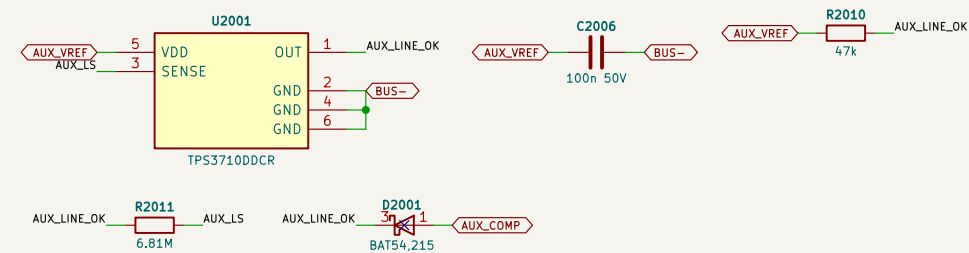
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



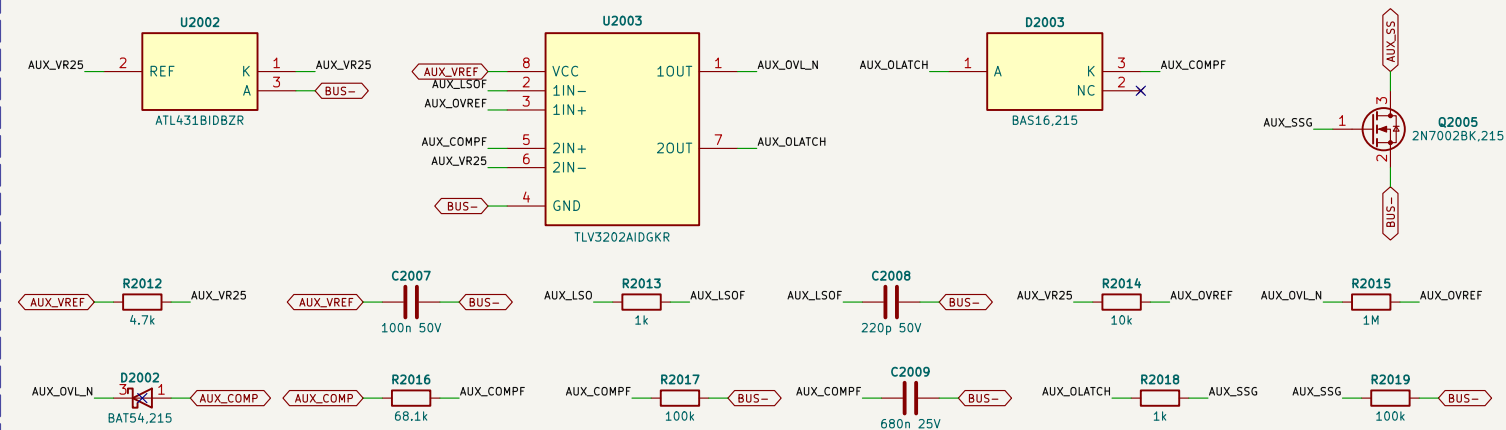
VREF high → start-up off (as AUX-HV).

LINE BROWN-IN



LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

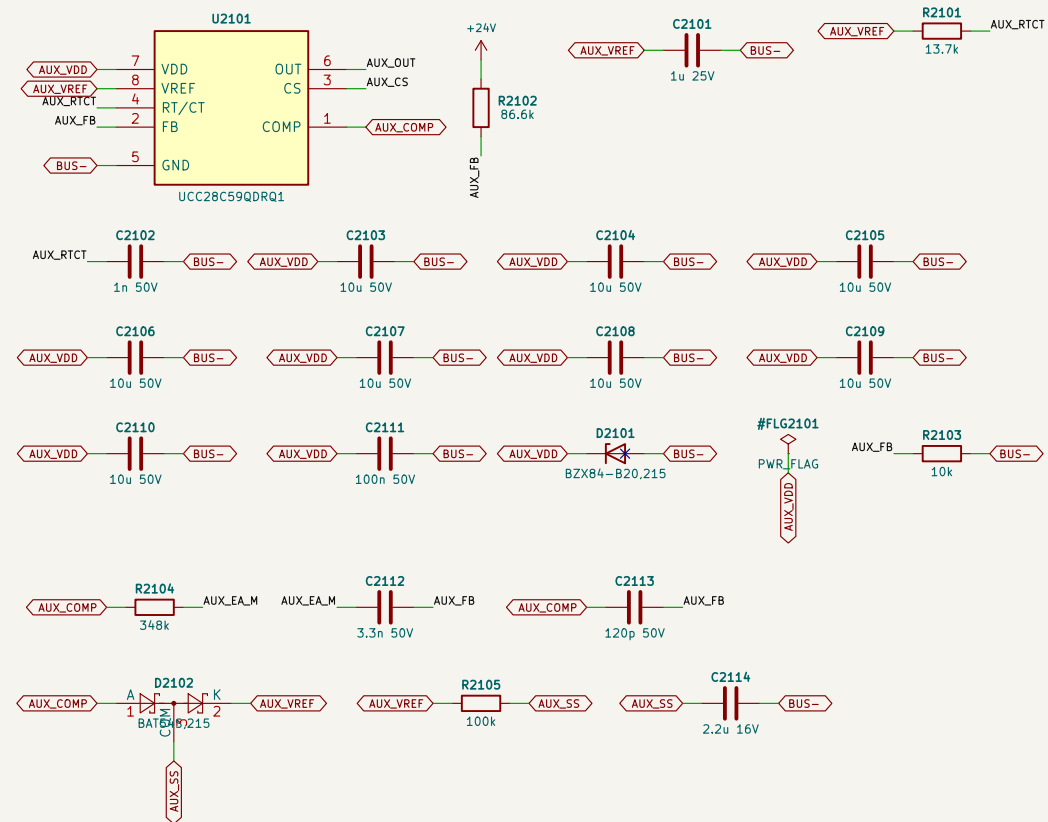
INPUT OV LOCKOUT AND OVERLOAD TIMER



Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 103 / 160 / 256 W (no trip at 75 W + 6 W pull-in); timer 35-80 ms -> latch -> SS crowbar -> hiccup.

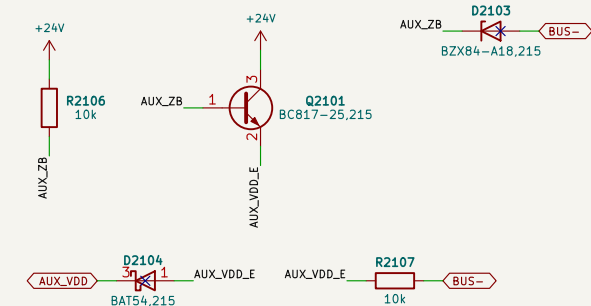
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 21 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /18 AUX75 start-up, brown-in, OV, timer/ File: 18_aux75_startup.kicad_sch		
Title: PV-PWR power board – AUX75 start-up, brown-in, OV, timer		
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6		Id: 21/23

PWM CONTROLLER AND ERROR AMPLIFIER



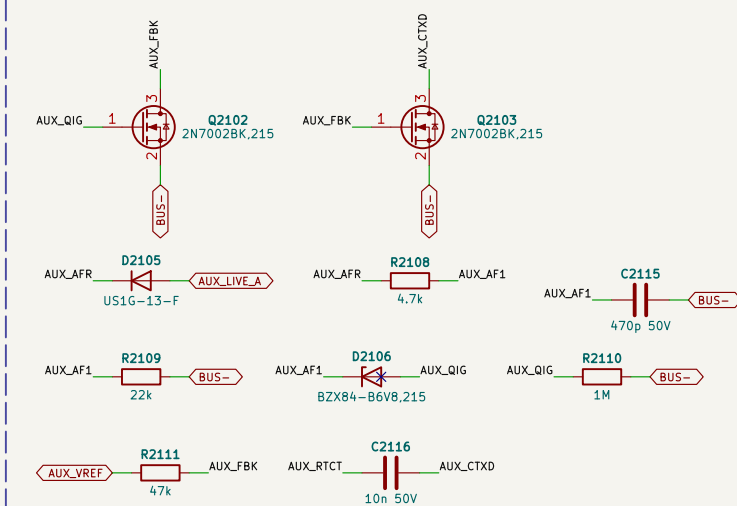
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 241–667 Hz, phase margin ≥ 58 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



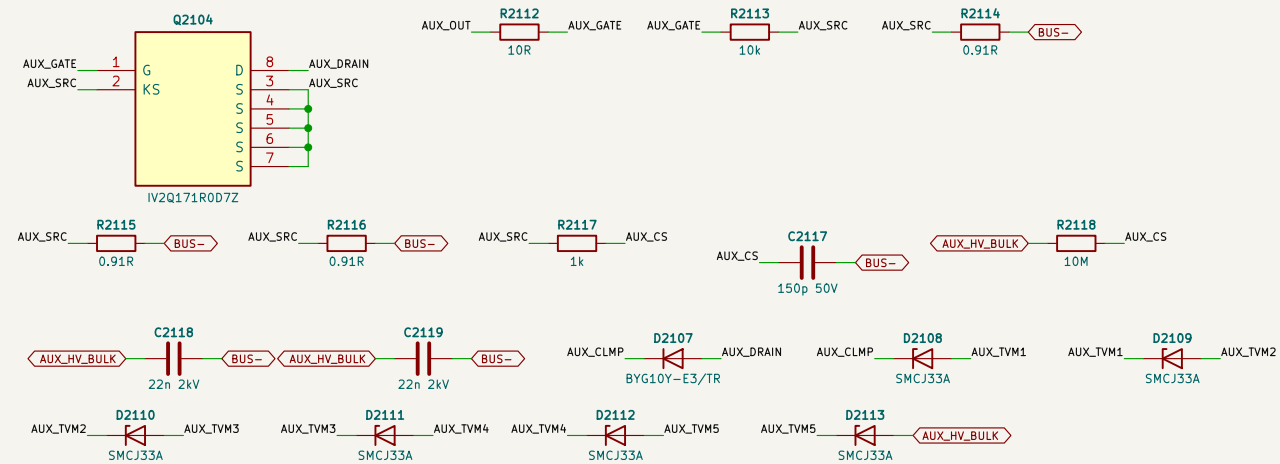
(30 W case, unchanged) follower Zener bias: VDD = gate drive 15.46–18.11 V (BZX84–A18, –40..85 °C)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX-03)



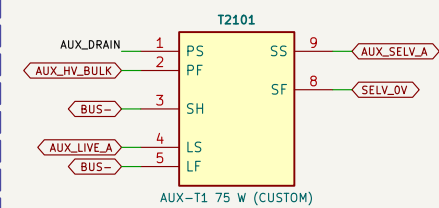
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output → fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



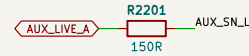
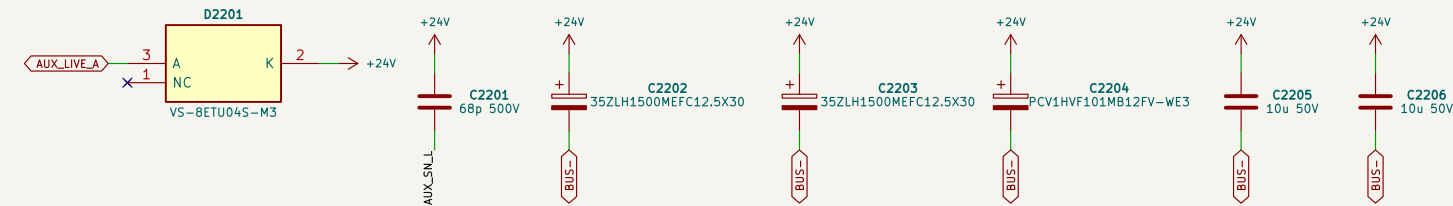
3 In parallel = 0.303333R; lowest limit 2.90 A delivers 103 W (>= 110 % of 75 W) at every input; highest 3.91 A with the line feed-forward (4.27 A without) line feed-forward HV_BULK -> CS (CRH2512); CS offset 101 mV at 1000 V cancels most of the 155-250 nV CS-JB delivery overshoot, so the highest current limit no longer rises with the input
Clamp: BYG10Y + 6 x SMC33A; V_DS 1316 / 1420 V at 1000 / 1100 V

AUX-T1 (CUSTOM – THE MODULE'S REINFORCED SUPPLY BARRIER B1)



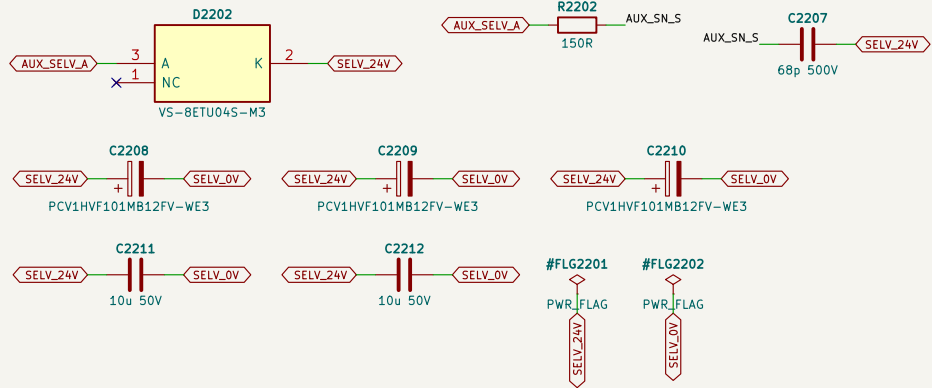
Lp 0.485 mH +/-7 %; shield and live winding on BUS-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

LIVE 24 V RECTIFIER, HOLD-UP



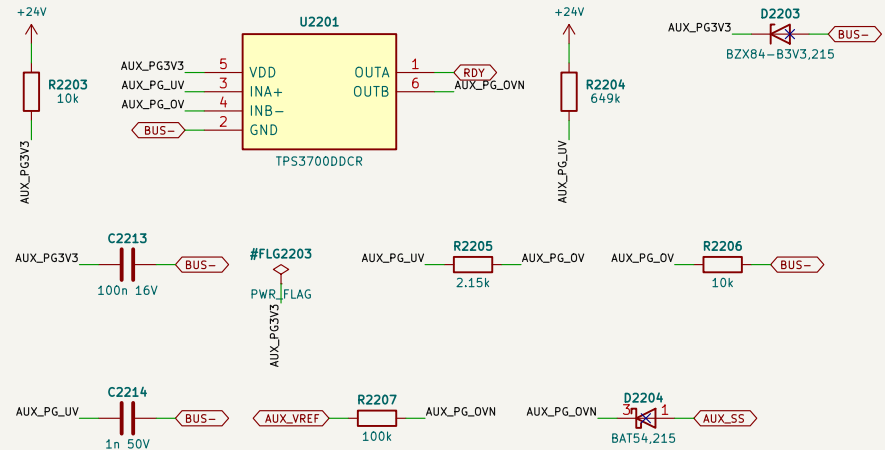
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)